

CHAPTER 6

RANDOM VARIABLES: MULTIVARIATE CASE

6.1 BIVARIATE DISTRIBUTIONS

The considerations of Section 5.2 can be extended to the case of several random variables analyzed at once, or equivalently, to the analysis of vector-valued random variables.

In the simplest case, we have a pair of random variables (X, Y) , that is, a pair of functions on the sample space $\mathcal{S}$.

■ **EXAMPLE 6.1**

Let the experiment consist of three tosses of a coin, and let X = number of heads in all three tosses and Y = number of tails in the last two tosses. The sample space $\mathcal{S}$ and corresponding values of X and Y are then

$\mathcal{S}$	X	Y	$\mathcal{S}$	X	Y
HHH	3	0	TTH	1	2
HHT	2	1	THT	1	1
HTH	2	1	HTT	1	1
THH	2	0	TTT	0	2

We can summarize all the possible values of (X, Y) and their probabilities in the following table:

X/Y	0	1	2
0	0	0	$\frac{1}{8}$
1	0	$\frac{2}{8}$	$\frac{1}{8}$
2	$\frac{1}{8}$	$\frac{2}{8}$	0
3	$\frac{1}{8}$	0	0

The entries in the table represent the corresponding probabilities. For instance, $P\{X = 2, Y = 1\} = 1/4$ was obtained by counting the number of points s of the sample space such that $X(s) = 2$ and $Y(s) = 1$ (there are two such points: HHT and HTH). To simplify the notation, we will be using commas to denote the intersection of events. Thus, we write $P\{X = x, Y = y\}$ instead of rather clumsy $P[\{X = x\} \cap \{Y = y\}]$.

In a natural way, this example leads to a definition of a *discrete* bivariate random variable.

Definition 6.1.1 We say that the pair (X, Y) of random variables has a *discrete* distribution if there exist finite or countable sets A and B such that

$$P\{(X, Y) \in A \times B\} = \sum_{x \in A, y \in B} P\{X = x, Y = y\} = 1. \quad \square$$

In Example 6.1, we have $A = \{0, 1, 2, 3\}$ and $B = \{0, 1, 2\}$. Obviously, $P\{X = x, Y = y\} = 0$ if any of the values x or y lie outside the set A (respectively, B), but it is also possible that $P\{X = x, Y = y\} = 0$ for some $x \in A$ and $y \in B$. For instance, in Example 6.1, we have $P\{X = 0, Y = 1\} = 0$. In other words, the values (x, y) that have positive probability can form a proper subset of $A \times B$.

The continuous bivariate distributions, as may be expected, are defined through their densities.

Definition 6.1.2 Random variables (X, Y) are *jointly continuous* (or have *joint continuous distribution*) if there exists a function $f(x, y)$ such that for every rectangle

$$C = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$$

with $-\infty \leq a < b \leq \infty, -\infty \leq c < d \leq \infty$ we have

$$P\{(X, Y) \in C\} = \int_C \int f(x, y) dx dy. \quad (6.1)$$

The function f is called *joint* or *bivariate* density of (X, Y) . $\square$

Some comments are in order here. First, as in the univariate case, the density is defined only up to sets of measure zero (e.g., single points or arcs on the plane).

Second, the obvious consequences of Definition 6.1.2 are

$$\int_{\mathbb{R}^2} \int f(x, y) dx dy = 1, \quad (6.2)$$

where R^2 is the plane, and

$$f(x, y) \geq 0 \quad \text{almost everywhere.} \quad (6.3)$$

These relations are obvious analogues of (5.22) and (5.25) for univariate densities.

The third comment is as follows: Formula (6.1) only covers rectangles. In the analogy with Theorem 5.2.5, it can be shown that probability P defined on the class of rectangles by (6.1) determines its unique extension to all sets in the plane that can be approximated through countable operations on rectangles, in particular to all figures that can be triangulated, as well as to circles, ellipses, and so on. In other words, formula (6.1) holds for a much wider class of sets C on the plane.

Conditions (6.2) and (6.3) allow us to use the Fubini theorem and replace the double integral (6.1) by the iterated integral, thus making actual integration possible. In particular, if C is the rectangle specified in Definition 6.1.2, then

$$\begin{aligned} P\{(X, Y) \in C\} &= \iint_C f(x, y) \, dx \, dy \\ &= \int_a^b \left[\int_c^d f(x, y) \, dy \right] dx = \int_c^d \left[\int_a^b f(x, y) \, dx \right] dy. \end{aligned}$$

In general, we have

$$\iint_C f(x, y) \, dx \, dy = \int_{C_2} \varphi(y) \, dy = \int_{C_1} \psi(x) \, dx.$$

Here C_1 and C_2 are “shadows” of C on the x -axis and y -axis, that is, $C_1 = \{x : (x, y) \in C\}$ for some y and $C_2 = \{y : (x, y) \in C\}$ for some x . Also

$$\psi(x) = \int_{C_x} f(x, y) \, dy, \quad \varphi(y) = \int_{C_y} f(x, y) \, dx$$

with $C_x = \{y : (x, y) \in C\}$ and $C_y = \{x : (x, y) \in C\}$ being sections of C at points x and y , respectively (see Figure 6.1). To understand well the difference between the double integral

$$\iint_{R^2} f(x, y) \, dx \, dy$$

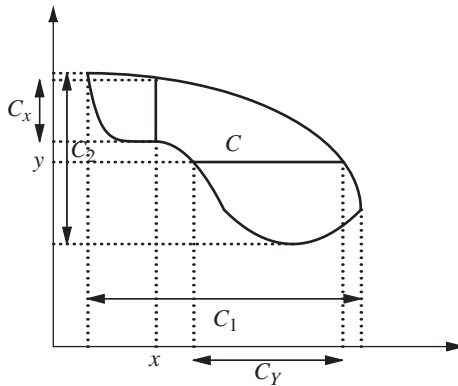


Figure 6.1 Shadows and sections of domain C of integration.

and iterated integrals

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) \, dx \, dy,$$

interpreted as

$$\int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} f(x, y) \, dx \right] dy \text{ and } \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} f(x, y) \, dy \right] dx,$$

the readers are advised to consult a good calculus text for the respective definitions. The Fubini theorem gives conditions under which all three integrals are equal (we will discuss these topics in Chapter 7). It is both easy and worthwhile to explain the issues involved here, using a simplified situation of a series (instead of integrals). Thus, $\sum_{n=1}^{\infty} a_n$ is *defined* as $\lim_{n \rightarrow \infty} \sum_{k=1}^n a_k$, provided that this limit (of a well-defined numerical sequence) exists. By the same argument, the double sum $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn}$ is defined unambiguously (this sum is an analogue of the iterated integral). However, the symbol $\sum_{m,n=1}^{\infty} a_{mn}$ (the analogue of a double integral) is *not* well defined, since it does not specify the order in which the two-dimensional array $\{a_{mn}\}$ is to be added. The point here is that the sum of an infinite sequence of numbers may depend on the order of summation. Thus, *some* assumptions about $\{a_{mn}\}$ are needed to make the last sum independent of the order of summation. Under these assumptions (e.g., nonnegativity) the double sum and both iterated sums are either all infinite or all equal to the same finite number.

A similar kind of difficulty appears in the case of double and iterated integrals and is resolved by the Fubini theorem.

We will now illustrate the calculation of probabilities by some examples.

■ EXAMPLE 6.2

The density $f(x, y)$ is given by the formula

$$f(x, y) = \begin{cases} cx(x+y) & \text{if } x \geq 0, y \geq 0, x+y \leq 1 \\ 0 & \text{otherwise.} \end{cases} \quad (6.4)$$

Find $P\{X \leq 1/2\}$.

SOLUTION. The first objective is to determine the constant c in formula (6.4). As usual, the normalizing condition (6.2) provides the key here:

$$\begin{aligned} 1 &= \int \int cx(x+y) \, dx \, dy = c \int_0^1 \left[\int_0^{1-y} c(x^2 + xy) \, dx \right] dy \\ &= c \int_0^1 \left(\frac{x^3}{3} + \frac{x^2}{2} y \Big|_0^{1-y} \right) dy = c \int_0^1 \left[\frac{(1-y)^3}{3} + \frac{(1-y)^2 y}{2} \right] dy = \frac{c}{8}, \end{aligned}$$

so $c = 8$.

Now, $P\{X \leq 1/2\}$ is the integral of the density $f(x, y)$ over the dark area in Figure 6.2. This time it is simpler to integrate over y first,

$$\begin{aligned} P\{X \leq 1/2\} &= \int_D \int 8x(x+y) \, dx \, dy = 8 \int_0^{1/2} \left[\int_0^{1-x} (x^2 + xy) \, dy \right] dx \\ &= 8 \int_0^{1/2} \left(x^2 y + x \frac{y^2}{2} \Big|_0^{1-x} \right) dx \\ &= 8 \int_0^{1/2} \left[x^2(1-x) + \frac{1}{2}x(1-x)^2 \right] dx = \frac{7}{16}. \end{aligned}$$

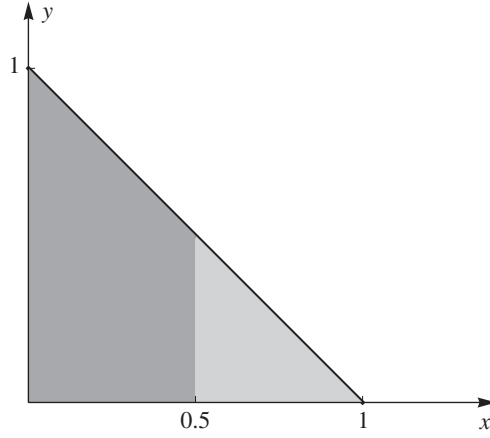


Figure 6.2 Support of density f and the set $\{X \leq 1/2\}$.

Although used not as often as in the univariate case, the cumulative distribution function (cdf) is still the most general way of specifying probabilities in the bivariate case, regardless of whether the distribution is discrete, continuous, or neither. We introduce the following definition:

Definition 6.1.3 The function of two real variables, defined as

$$F(x, y) = P\{X \leq x, Y \leq y\}$$

is called the *cdf* of the pair (X, Y) of random variables. $\square$

The relation between cdf and the probability mass function in the discrete case, or the density function in the continuous case, is similar to those for univariate random variables:

$$F(x, y) = \sum_{x_i \leq x, y_j \leq y} P\{X = x_i, Y = y_j\}$$

in the discrete case, and

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) \, dv \, du \quad (6.5)$$

in the continuous case. In particular, from (6.5) it follows that for almost all (x, y) , we have

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}.$$

This brings us to the following analogue of Theorem 5.2.2:

Theorem 6.1.1 Every bivariate cdf $F(x, y)$ has the following properties:

- (a) $\lim_{x, y \rightarrow +\infty} F(x, y) = 1$.
- (b) For every y , $F(x, y)$ is nondecreasing and continuous from the right in x .
- (c) For every x , $F(x, y)$ is nondecreasing and continuous from the right in y .
- (d) $\lim_{x \rightarrow -\infty} F(x, y) = 0$ for every y , and $\lim_{y \rightarrow -\infty} F(x, y) = 0$ for every x .
- (e) For all $x_1 < x_2$ and $y_1 < y_2$,

$$F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1) \geq 0. \quad (6.6)$$

We omit the proof here, leaving the proofs of some of the properties as exercises. The following comments, however, are important. While conditions (a)–(d) are direct analogue of the properties of univariate cdf's, condition (e) has no counterpart. The question therefore arises whether conditions (a)–(d) alone characterize bivariate cdf's. In other words, is every function satisfying (a)–(d) a cdf of some pair (X, Y) of random variables? The answer is negative, as shown by the following example:

■ EXAMPLE 6.3

Let $F(x, y)$ be defined by the formula

$$F(x, y) = \begin{cases} 1 & \text{if } x + y \geq 0 \\ 0 & \text{if } x + y < 0. \end{cases} \quad (6.7)$$

It is easy to check that function (6.7) satisfies conditions (a)–(d) given above. Suppose that $F(x, y)$ is the cdf of a pair of random variables, and let us compute the probability (see Figure 6.3), $P\{-1 < X \leq 2, -1 < Y \leq 2\}$.

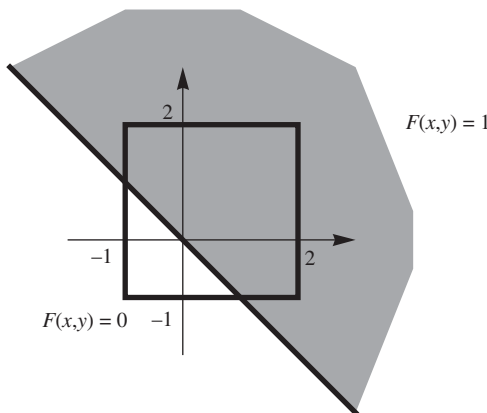


Figure 6.3 A function that is not a cdf but satisfies (a)–(d).

Since $F(x, y)$ is the probability of the pair (X, Y) taking a value “southwest” of (x, y) [i.e., to the left and below (x, y)], we have

$$F(2, 2) - F(-1, 2) - F(2, -1) + F(-1, -1) = 1 - 1 - 1 + 0 = -1,$$

so that $F(x, y)$ cannot be a cdf.

Condition (e) is necessary, since the left-hand side of (6.6) equals $P\{x_1 < X \leq x_2, y_1 < Y \leq y_2\}$ and hence must be nonnegative.

It turns out (we omit the proof here) that (a)–(e) characterize a bivariate cdf. This means that any function satisfying (a)–(e) is a cdf of some pair of random variables.

PROBLEMS

- 6.1.1** A regular die is tossed twice. Find: **(i)** The joint distribution of variables X = the total of both outcomes and Y = the best of the two outcomes. **(ii)** $P(X \leq 8, Y \leq 5)$; $P(X = 9, Y \leq 2)$, $P(4 \leq X \leq 7, 1 \leq Y \leq 3)$. **(iii)** $P(Y = 3 | X = 4)$, $P(Y < 6 | X = 7)$, $P(4 < Y \leq 6 | X \leq 8)$.

6.1.2 Let X, Y have the joint distribution given by the following table:

X/Y	2	3	4
0	$\frac{1}{48}$	0	b
1	0	$\frac{5}{48}$	$\frac{8}{48}$
2	a	0	$\frac{11}{48}$
3	$\frac{5}{48}$	$\frac{12}{48}$	0

(i) Find a and b if it is known that $P(X = Y) = 1/3$. (ii) Find $P(XY = 0)$. (iii) If F is the cdf of (X, Y) , find $F(-1.5, 3)$, $F(0.7, 2.11)$, and $F(1.5, 18)$.

6.1.3 A regular die is tossed twice. Let X be the number of times that 1 came up, and Y be the number of times 2 came up. Find: (i) The joint distribution of X and Y . (ii) The correlation coefficient between events $\{X = 1\}$ and $\{Y = 2\}$. [Hint: See formula (4.17) in Definition 4.5.2.]

6.1.4 Let the joint cdf of random variables X, Y be $F(x, y) = (1/48)xy(x + 2y)$ for $0 \leq x \leq 2, 0 \leq y \leq 3$. Find the density $f(x, y)$.

6.1.5 The joint density of X and Y is $f(x, y) = y^2(xy^3 + 1)$ on the rectangle $0 \leq x \leq k, 0 \leq y \leq 1$. Find: (i) k . (ii) $P(X \leq Y)$.

6.1.6 Assume that X, Y have density $f(x, y) = x + y$ for $0 \leq x \leq 1$ and $0 \leq y \leq 1$, and $f(x, y) = 0$, otherwise. Find $P\{Y \leq \sqrt[3]{X}\}$.

6.1.7 Assume that (X, Y) have the joint density $f(x, y) = cxy^2$ for $0 \leq x \leq 1, 0 \leq y \leq 1$, and $f(x, y) = 0$, otherwise. Find: (i) c . (ii) $P\{X^2 \leq Y \leq X\}$. (iii) The cdf of (X, Y) .

6.1.8 Let the joint density of random variables X, Y be $f(x, y) = cx^3y^2$ for $0 \leq x \leq 1, x^2 \leq y \leq 1$ and $f(x, y) = 0$, otherwise. Find $P(X < Y)$.

6.1.9 Variables X and Y have the joint density $f(x, y) = 1/y$ for $0 < x < y < 1$ and $f(x, y) = 0$, otherwise. Find: (i) $P(X + Y > 0.5)$. (ii) $P(Y > 2X)$.

6.2 MARGINAL DISTRIBUTIONS; INDEPENDENCE

One can naturally expect that the bivariate distribution (in the form of cdf, joint density, or probability mass function, as the case might be) contains more information than the univariate distributions of X and Y separately. Given a bivariate distribution, we are able to recover both univariate distributions of X and of Y , but not conversely.

We begin with the case of discrete bivariate distributions. Let $A = \{x_1, x_2, \dots\}$ and $B = \{y_1, y_2, \dots\}$ be the sets of possible values of X and Y , respectively, and let

$$p_{ij} = P\{X = x_i, Y = y_j\}. \quad (6.8)$$

Our objective is to express the distributions of X and of Y through p_{ij} . Since the events $\{Y = y_1\}, \{Y = y_2\}, \dots$ form a partition (i.e., in the sense defined in Chapter 4, these events are mutually exclusive and one of them has to occur), we may write for every i ,

$$\{X = x_i\} = \bigcup_j [\{X = x_i\} \cap \{Y = y_j\}] = \bigcup_j \{X = x_i, Y = y_j\}.$$

Since the events on the right-hand side are disjoint, we have

$$P\{X = x_i\} = \sum_j P\{X = x_i, Y = y_j\} = \sum_j p_{ij}.$$

In a similar way, $P\{Y = y_j\} = \sum_i p_{ij}$. We now introduce

Definition 6.2.1 Given the joint distribution (6.8), the distributions of X alone and Y alone, calculated from the formulas

$$P\{X = x_i\} = \sum_j p_{ij} = p_{i+}, \quad P\{Y = y_j\} = \sum_i p_{ij} = p_{+j},$$

will be referred to as *marginal* distributions. □

If we think of numbers p_{ij} as arranged into a matrix, then $P\{X = x_i\}$ and $P\{Y = y_j\}$ are sums of its corresponding rows (or columns). Since the sum of all p_{ij} equals 1, both marginal distributions satisfy the condition that the sum of their probabilities equals 1.

■ EXAMPLE 6.4

In Example 6.1, we considered three tosses of a coin, with X being the number of heads in all three tosses, and Y being the number of tails in the last two tosses. The joint distribution of X and Y is summarized by the following table:

X/Y	0	1	2	X
0	0	0	$\frac{1}{8}$	$\frac{1}{8}$
1	0	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{3}{8}$
2	$\frac{1}{8}$	$\frac{2}{8}$	0	$\frac{3}{8}$
3	$\frac{1}{8}$	0	0	$\frac{1}{8}$
Y	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	1

In the margins, we have the row sums and column sums. The distribution of X is

$$P\{X = 0\} = P\{X = 3\} = \frac{1}{8}, \quad P\{X = 1\} = P\{X = 2\} = \frac{3}{8},$$

while the distribution of Y is

$$P\{Y = 0\} = P\{Y = 2\} = \frac{1}{4}, \quad P\{Y = 1\} = \frac{1}{2}.$$

The adjective *marginal* refers to the way in which the distribution was obtained; it implies nothing about the properties of the distribution. The definition of marginal distribution for continuous random variables is analogous, with summation replaced by integration:

Definition 6.2.2 If (X, Y) is a pair of continuous random variables with bivariate density $f(x, y)$, then the functions

$$f_1(x) = \int_{-\infty}^{+\infty} f(x, y) dy \quad \text{and} \quad f_2(y) = \int_{-\infty}^{+\infty} f(x, y) dx$$

are called the *marginal densities* of variables X and Y , respectively. □

The justification of Definition 6.2.2 consists of two parts. First, we need to show that f_1 and f_2 are densities, meaning that they are nonnegative for almost all arguments (i.e., for all arguments, except a set of arguments of measure zero) and that f_1 and f_2 integrate to 1. These properties are immediate consequences of the fact that $f(x, y) \geq 0$ except possibly on a set of measure zero. Consequently, using the Fubini theorem, we can write

$$\begin{aligned} 1 &= \int \int f(x, y) dx dy = \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} f(x, y) dx \right] dy \\ &= \int_{-\infty}^{+\infty} f_2(y) dy, \end{aligned}$$

and similarly for f_1 .

The second part of justification of Definition 6.2.2 consists of showing that, for instance,

$$P\{a \leq X \leq b\} = \int_a^b f_1(x) dx.$$

Here we can write

$$\begin{aligned} \int_a^b f_1(x) dx &= \int_a^b \left[\int_{-\infty}^{+\infty} f(x, y) dy \right] dx \\ &= \int \int_{a \leq x \leq b, -\infty < y < +\infty} f(x, y) dx dy \\ &= P\{a \leq X \leq b, -\infty < Y < +\infty\} = P\{a \leq X \leq b\}. \end{aligned}$$

■ EXAMPLE 6.5

A man shoots at a circular target. Assume that with his skills he is certain to hit the target and that the probability of hitting a particular part of the target is proportional to the area of this part. If X and Y be the horizontal and vertical distances from the center of the target to the point of impact, find the distribution of X .

SOLUTION. Without a loss of generality, we can introduce the coordinate system with the origin in the center of the target. Let us assume that its radius is R . Then the target is described as $x^2 + y^2 \leq R^2$. From the conditions of the problem, it is seen that the density of the point of impact is constant on the target so that

$$f(x, y) = \begin{cases} c & \text{if } x^2 + y^2 \leq R^2 \\ 0 & \text{otherwise.} \end{cases}$$

Next, the condition $\int \int_{R^2} f(x, y) dx dy = 1$ implies that c must be the reciprocal of the area of the target; hence, $c = 1/\pi R^2$.

If $|x| > R$, then $f(x, y) = 0$ for all y (see Figure 6.4). For $|x| \leq R$, we have $f(x, y) = 0$ when $x^2 + y^2 > R^2$ (hence if $|y| > \sqrt{R^2 - x^2}$), and $f(x, y) = 1/\pi R^2$ when $x^2 + y^2 \leq R^2$ (so $|y| \leq \sqrt{R^2 - x^2}$). Consequently, for $|x| \leq R$,

$$f_1(x) = \int_{-\sqrt{R^2 - x^2}}^{\sqrt{R^2 - x^2}} \frac{1}{\pi R^2} dy = \frac{2\sqrt{R^2 - x^2}}{\pi R^2}.$$

The density $f_1(x)$ has the shape given in Figure 6.4.

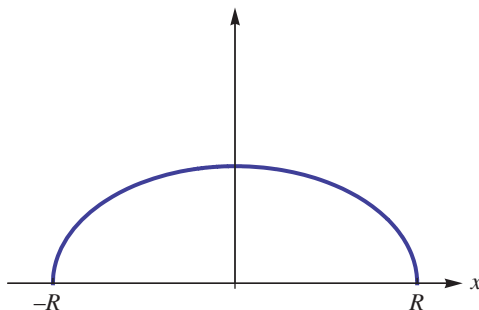


Figure 6.4 Marginal density.

At this point, we need to comment on notation. The symbols $f_1(x)$ and $f_2(y)$ for marginal densities of X and Y are used in the literature together with more readable symbols such as $f_X(x)$ and $f_Y(y)$, with the name of the variable appearing as a subscript. This latter system becomes cumbersome when the variables are labeled X_1 and X_2 instead of X and Y , since logically one should then use $f_{X_1}(x)$ and $f_{X_2}(y)$. We will deliberately avoid keeping rigidly to any fixed system of notation, and use whichever notation appears most natural in a given instance. Another problem concerns the use of an argument in the density or cdf. It appears natural to label the argument x in density or cdf of X , label it y in density of Y , and so on. However, it is *not possible* to use such notation consistently: Indeed, if $F(x)$ and $f(x)$ are to be used as the cdf and density of X , then we have $F(x) = \int_{-\infty}^x f(u) du$, and in the integrand we can use almost any symbol *except* x , since the symbol $\int_{-\infty}^x f(x) dx$ is unacceptably ambiguous.

We will now define the concept of independence of two random variables. We recall that the independence of two events A and B is defined by the product rule $P(A \cap B) = P(A)P(B)$ (see Chapter 4). A random variable X allows us to define events of the form $\{X \in A\}$, where A is some set of real numbers, and the same is true for the random variable Y . It seems natural to require that these events be independent for independent random variables X and Y .

Definition 6.2.3 (Intention of the Concept) We say that random variables X and Y are independent if events $\{X \in A\}$ and $\{Y \in B\}$ are independent for all sets A, B , that is, if

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}. \quad (6.9)$$

□

This definition, as spelled out in its label, concerns the “final effect” of the concept: independence of *every pair* of events in a very large class of such pairs. Clearly, using this definition to check independence would be very difficult if we were to verify the product rule (6.9) for every pair A, B . Consequently, it becomes necessary to find a condition that is verifiable and strong enough to imply independence in the sense of Definition 6.2.3. It turns out that it is sufficient to require only a seemingly weaker condition.

Definition 6.2.4 (Verifiable Definition of Independence) The random variables X and Y are said to be *independent* if their joint and marginal cdf's satisfy the following condition: For every x, y we have

$$F(x, y) = F_1(x)F_2(y), \quad (6.10)$$

where $F_1(x)$ and $F_2(y)$ are marginal cdf's of X and Y , respectively.

□

In the case of discrete random variables with p_{ij} being the probability $P\{X = x_i, Y = y_j\}$, condition (6.10) is implied by

$$p_{ij} = p_{i+} p_{+j} \quad \text{for every } i, j. \quad (6.11)$$

For continuous random variables, the condition implying (6.10) calls for marginal densities f_1 and f_2 to be such that

$$f(x, y) = f_1(x) f_2(y) \quad (6.12)$$

for all x, y , except possibly for a set of points (x, y) of measure zero.

The intuition behind the concept of independence of random variables (analogous to the intuition behind the concept of independence of events) is that the information about the value of one of them provides no information about the value of the other. Random variables for which independence conditions are not satisfied will be called dependent.

■ EXAMPLE 6.6

Let us go back to Example 6.1. In the table of joint and marginal Distribution, we had $P\{X = 3, Y = 2\} = 0$, while $P\{X = 3\} \times P\{Y = 2\} = \frac{1}{8} \times \frac{1}{4} \neq 0$. This means that X and Y are dependent random variables.

■ EXAMPLE 6.7

Let us consider again the case of three tosses of a coin. Let U be the number of heads in the first two tosses, and let V be the number of tails in the last toss. The sample points are

S	U	V	S	U	V
HHH	2	0	HTT	1	1
HHT	2	1	THT	1	1
HTH	1	0	TTH	0	0
THH	1	0	TTT	0	1

The joint and marginal probability distributions are

U/V	0	1	U
0	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
1	$\frac{2}{8}$	$\frac{2}{8}$	$\frac{1}{2}$
2	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
V	$\frac{1}{2}$	$\frac{1}{2}$	1

A direct check shows that we have

$$P\{U = x, V = y\} = P\{U = x\} \times P\{V = y\}$$

for every cell (x, y) in the table above. This shows that U and V are independent.

Verification that two random variables are independent requires checking the multiplicative property (6.11) or (6.12) for *all* x and y . In the last example, this required comparing joint probabilities with the products of probabilities in marginal distributions for all six cells

in the joint distribution table. Such a direct verification is not feasible except for discrete variables with small sets of possible values. To handle more complicated cases, we typically must have some algebraic formula for the joint distribution from which we can calculate the marginal distributions and verify the product rule algebraically. On the other hand, to show that two variables are *dependent* (i.e., that they are *not independent*), it is enough to find *one* pair (x, y) for which the product rule does not hold.

A practical consequence here is that it is generally worthwhile to try to determine based on the meaning of the variables in question, whether or not we can expect them to be independent. This determines the strategy: Do we aim at checking that variables are independent, or do we aim at showing that they are dependent? These two goals may require somewhat different types of technique.

To illustrate the point, in Example 6.6, we could have expected the variables to be dependent: the more tails in the last two tosses (Y), the lower one can expect the total number of heads (X) to be. In particular, we may be able to find a “pure exclusion,” such as the fact that one cannot have $X = 3$ and $Y = 2$ simultaneously, while separately each of these events has positive probability.

On the other hand, in Example 6.7 we could have expected U and V to be independent, since the values of U and V were determined by nonoverlapping sets of tosses. In particular, the following criterion is useful in showing that two variables are *not independent*.

Theorem 6.2.1 *If the table of joint probabilities for (X, Y) contains a zero entry, then X and Y are dependent.*

Proof: Suppose that $P\{X = x_0, Y = y_0\} = 0$. Since x_0 and y_0 have positive probabilities, $P(X = x_0) \times P(Y = y_0)$ is positive and (6.11) is not satisfied. $\square$

This criterion provides a quick “visual” test for lack of independence. Of course, it works only in one direction: if there are no zeros in the table of joint distribution, the variables may or may not be dependent, and further checking is necessary.

We will now consider some examples of applications of Definition 6.2.4 for the case of continuous random variables.

■ EXAMPLE 6.8

The joint density of random variables X and Y is

$$f(x, y) = \begin{cases} cx^n e^{-\alpha x - \beta y} & \text{for } x > 0, y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

The question is whether these random variables are independent.

SOLUTION. It is interesting that the answer here does not require any calculations. In particular, we do not need to calculate c . We can simply represent the joint density on the positive quadrant as

$$f(x, y) = c_1 x^n e^{-\alpha x} \times c_2 e^{-\beta y} \quad (6.13)$$

with c_1 and c_2 such that

$$c_1 \int_0^\infty x^n e^{-\alpha x} dx = 1 \quad \text{and} \quad c_2 \int_0^\infty e^{-\beta y} dy = 1$$

(then automatically $c = c_1 c_2$). The two factors on the right of (6.13) must be marginal densities (why?), and we showed independence of X and Y . However, for the joint density $f(x, y)$ to factor into the product of two functions, each depending on one variable only, is not enough to warrant the independence of variables X and Y , since the marginal densities depend on the shape of the support of the joint density (i.e., the set of points where $f(x, y)$ is positive). To see it, consider the following example.

■ EXAMPLE 6.9

Let the joint density be

$$f(x, y) = \begin{cases} cxy^2 & \text{if } x \geq 0, y \geq 0, x + y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

At first glance, the situation here is similar to that in Example 6.8, but it is not. The marginal distributions are *not* $c_1 x$ and $c_2 y^2$ for any c_1 and c_2 . Indeed, (see Figure 6.2 for limits of integration):

$$\begin{aligned} f_1(x) &= \int_{-\infty}^{+\infty} f(x, y) dy = \int_{-\infty}^0 0 dy + \int_0^{1-x} cxy^2 dy + \int_{1-x}^{\infty} 0 dy \\ &= \frac{cx(1-x)^3}{3} \end{aligned}$$

for $0 < x < 1$, and $f_1(x) = 0$ for $x > 1$ and $x < 0$. Similarly, $f_2(y) = 0$ if $y > 1$ or $y < 0$, and for $0 < y < 1$,

$$f_2(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \frac{Cy^2(1-y)^2}{2}.$$

Since $f(x, y) \neq f_1(x)f_2(y)$, the random variables are dependent.

We can now formulate

Theorem 6.2.2 Assume that the joint density $f(x, y)$ is continuous, and let $f(x, y) > 0$ for $(x, y) \in A$. If A is not a Cartesian product $A = A_X \times A_Y$, then variables X and Y are dependent.

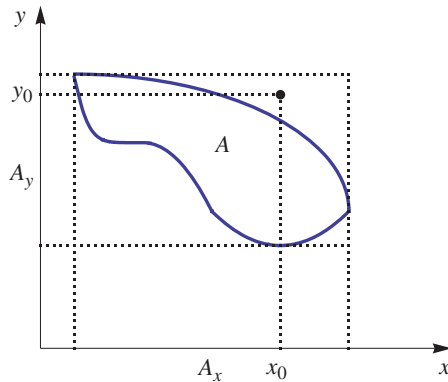


Figure 6.5 Condition for dependence.

Proof: The marginal densities f_1 and f_2 are strictly positive in the “shadows” A_X and A_Y of the set A on the x -axis and y -axis (see Figure 6.5). If A is not equal to the Cartesian product $A_X \times A_Y$, then there exists a point (x_0, y_0) with $x_0 \in A_X, y_0 \in A_Y$, and $(x_0, y_0) \notin A$. If $x_0 \in A_X$, then there exists y^* such that $(x_0, y^*) \in A$, and therefore $f(x_0, y^*) > 0$. Since f is continuous, it must be positive in some neighborhood of (x_0, y^*) , and therefore, $f_1(x_0) = \int f(x_0, y) dy > 0$. Similarly $f_2(y_0) > 0$. Consequently, $0 = f(x_0, y_0) \neq f_1(x_0)f_2(y_0) > 0$. The continuity of $f(x, y)$, and hence of $f_1(x)f_2(y)$, implies now that $f(x, y) \neq f_1(x)f_2(y)$ on a set of a positive probability. $\square$

While the continuity of $f(x, y)$ is sufficient for the criterion given by Theorem 6.2.2, it is not necessary. However, it ought to be mentioned here that this criterion should be applied with some caution if f is not continuous, since $f(x, y)$ is defined only up to sets of measure zero.

■ EXAMPLE 6.10

In Example 6.9, the set A is a triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$, so it can be inferred without any calculations that X and Y are dependent.

At the end of this section, let us make the following important remark. The formulas (6.9) through (6.12) from Definitions 6.2.3 and 6.2.4 can be used in two ways. The first is as illustrated so far: to determine whether or not two variables are independent. However, a more frequent use of these formulas is to find the joint distribution (joint density, joint cdf, etc.) of independent variables X and Y .

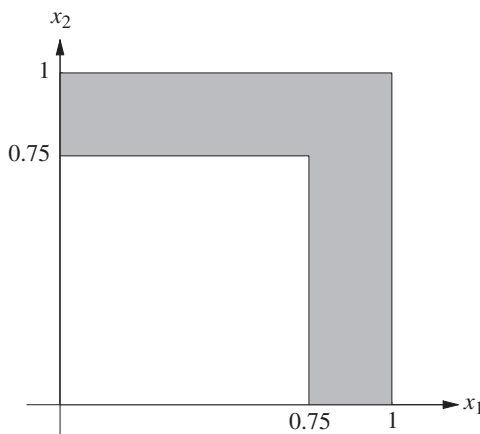


Figure 6.6 Probability of better of two attempts exceeding 0.75.

■ EXAMPLE 6.11

An athlete makes two attempts at some goal. His performance X at the first attempt, measured on the scale from 0 to 1, is a random variable with density $f_1(x) = 12x^2(1 - x)$. His performance Y at the second attempt is independent of X , and generally tends to be lower; its density is $f_2(y) = 6y(1 - y), 0 \leq y \leq 1$. What is the probability that level 0.75 is exceeded in the better of the two attempts?

SOLUTION. The joint density of (X, Y) is

$$f(x, y) = \begin{cases} 72x^2(1-x)y(1-y) & \text{for } 0 \leq x \leq 1, \ 0 \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

The required probability is obtained as the integral of $f(x, y)$ over the shaded area in Figure 6.6 hence equals

$$1 - \int_0^{0.75} \int_0^{0.75} 72x_1^2(1-x_1)x_2(1-x_2) dx_1 dx_2 = 0.3771.$$

PROBLEMS

- 6.2.1** The joint probability function of variables X and Y is $f(x, y) = c|x - y|$, $x = 0, 1, 2, 3$, and $y = 0, 1, 2$. Find: **(i)** c . **(ii)** $P(X = Y)$. **(iii)** $P(X > Y)$. **(iv)** $P(|X - 1| \leq 1)$. **(v)** $P(X + Y \leq 3)$.
- 6.2.2** Two cards are drawn at random from the ordinary deck of cards. Let X be the number of aces and let Y be the number of hearts obtained. **(i)** Find the joint probability function of (X, Y) . **(ii)** Find the marginal distribution of X and Y . **(iii)** Are X and Y independent?
- 6.2.3** An urn contains five balls, two of them red and three green. Three balls are drawn without replacement. Let X and Y denote the number of red (X) and green (Y) balls drawn. **(i)** Find the joint distribution of (X, Y) . **(ii)** Find the marginal distributions of X and Y . **(iii)** Are X and Y independent? **(iv)** Find the joint distribution of X and $Z = Y - X$.
- 6.2.4** A box contains three coconut candies, five hazelnut chocolates, and two peanut butter chocolates. A sample of four sweets is chosen from the box. Let X, Y , and Z be the number of coconut candies, hazelnut chocolates, and peanut butter chocolates in the sample, respectively. **(i)** Find the joint distribution of (X, Y) . **(ii)** Find the marginal distributions of X, Y , and Z . **(iii)** Are X and Y independent? Are X and Z independent?
- 6.2.5** Let X and Y have the joint distribution $P\{X = x, Y = y\} = c\lambda^{x+y}/(x!y!)$ for $x = 0, 1, \dots, y = 0, 1, \dots$, and $\lambda > 0$. **(i)** Find c . **(ii)** Find the marginal distribution of X . **(iii)** Are X and Y independent?
- 6.2.6** Random variables X and Y have joint distribution given by the following table:

X/Y	1	2	3
1	$\frac{1}{3}$	a	$\frac{1}{6}$
2	b	$\frac{1}{4}$	c

Show that X and Y are dependent, regardless of values a, b , and c .

- 6.2.7** Random variables have joint distribution given by the table

X/Y	1	2	3
1	a	$2a$	$3a$
2	b	c	d

Find a, b, c, d if X, Y are independent, and $P(X = 2) = 2P(X = 1)$.

- 6.2.8** Consider a system consisting of three components connected as in Figure 6.7. Let Y_1, Y_2, Y_3 be independent lifetimes of components 1, 2, and 3, respectively, each with EXP(α) distribution. If T is the lifetime of the whole system, find: **(i)** The cdf of T . **(ii)** The hazard function of T .

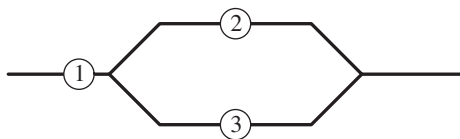


Figure 6.7 Three-component system.

- 6.2.9** Let variables X and Y have joint density $f(x, y) = 3(x + y)$ for $x > 0, y > 0, x + y < 1$ and 0 otherwise. Obtain: **(i)** The density and the cdf of the marginal distribution of X . **(ii)** $P(X + Y > 0.5 | X < 0.5)$.
- 6.2.10** Let (X, Y) have the distribution given by the table

X/Y	3	4	5	6
1	$\frac{1}{24}$	$\frac{1}{2}$	$\frac{1}{24}$	$\frac{1}{24}$
2	$\frac{1}{24}$	0	$\frac{1}{12}$	$\frac{1}{12}$
3	0	$\frac{1}{12}$	$\frac{1}{12}$	0

Find the probability distribution of independent random variables (X', Y') such that X' and Y' have the same marginals as X and Y .

- 6.2.11** Let T_1, T_2 be independent random variables with hazard functions $h_1(t)$ and $h_2(t)$, respectively. **(i)** Show that the variable with the hazard function $h(t) = h_1(t) + h_2(t)$ has the same distribution as $\min(T_1, T_2)$. **(ii)** Express $P(T_1 < T_2)$ through h_1 and h_2 .
- 6.2.12** Let variables X and Y be independent, with distributions GAM(1, 1) and GAM(2, 1), respectively. Find: **(i)** $P(X - Y > 0)$. **(ii)** $P(|X - Y| < 1)$.
- 6.2.13** Let X and Y be the lifetimes of two components of a machine. Their joint distribution is given by the density $f(x, y) = xe^{-x(1+y)}$ for $x > 0, y > 0$ and zero, otherwise. **(i)** Find $P(X \geq 5)$. **(ii)** Find the probability that $\max(X, Y) > 2$. **(iii)** Check the independence of X and Y using their marginal densities.
- 6.2.14** Random variables X and Y have joint density

$$f(x, y) = \begin{cases} k(ax + by) & 0 < x < 1, 0 < y < 2 \\ 0 & \text{otherwise,} \end{cases}$$

where $a > 0, b > 0$. Find: **(i)** k (as a function of a and b). **(ii)** The marginal distributions of X and Y . **(iii)** The cdf of (X, Y) .

- 6.2.15** Let X and Y be the time (in hours) that a customer spends at a service counter and the time he is actually being served ($Y < X$). The joint distribution of X and Y has density $f(x, y) = 4e^{-2x}$ for $0 < y < x$. Find: **(i)** The marginal distributions of X and Y . **(ii)** The cdf of (X, Y) .
- 6.2.16** Students have to randomly select a point inside a circular region with radius R . One of them first samples the direction from the center of the region according to a uniform distribution on $[0^\circ, 360^\circ]$, and then samples the distance from the center according to $U[0, R]$. Find: **(i)** The density $f(x, y)$ of the chosen points in (x, y) coordinates. **(ii)** The marginal distribution of X .
- 6.2.17** Assume that X and Y are independent random variables with $\text{EXP}(a)$ and $\text{EXP}(b)$ distributions, respectively. Assume that it is not possible to observe both X and Y but that one can observe

$$U = \min(X, Y) \text{ and } Z = \begin{cases} 1 & \text{if } X < Y \\ 0 & \text{otherwise.} \end{cases}$$

This kind of situation, called *censoring*, occurs often in medicine and engineering. For instance, X and Y may be the times of death due to heart failure (X) and death due to other causes (Y). Then U is the observed lifetime, and Z is the indicator of the cause of death. **(i)** Find the joint distribution of (U, Z) . **(ii)** Prove that U and Z are independent. (*Hint:* Note that U is continuous, Z is discrete. It suffices to show that $P\{U \leq u | Z = i\}$ is independent of i , $i = 0, 1$.)

- 6.2.18** Let X, Y be independent, continuous random variables with a symmetric (but possibly different) distribution around 0. Show that Y/X and $Y/|X|$ have the same distribution. (*Hint:* Compare the cdf's of $W = X/Y$ and $V = X/|Y|$.)
- 6.2.19** Let X and Y have distribution uniform in the shape of the letter Y (see Figure 6.8). Identify the shapes of the marginal densities of X and Y in Figure 6.9.

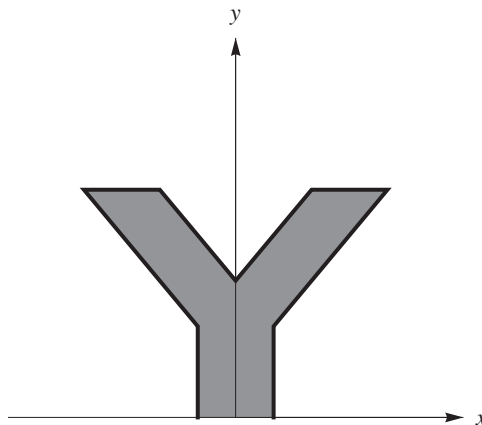


Figure 6.8 Joint distribution of X and Y .

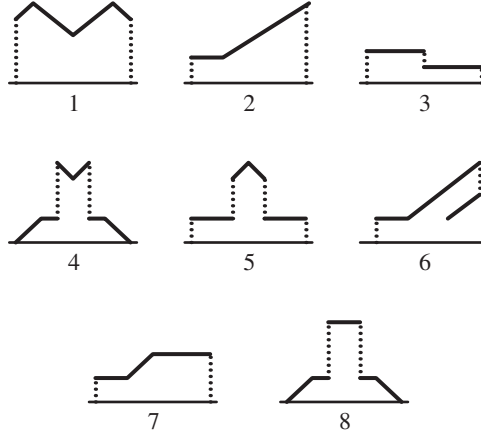


Figure 6.9 Options for marginal densities of X and Y .

6.3 CONDITIONAL DISTRIBUTIONS

A natural question concerning two random variables is how to handle situations where the information about the value of one of the variables affects the distribution of the other.

In symbols, the objective now is to determine the probabilities of the form

$$P(X \in Q_1 | Y = y) \text{ or } P(Y \in Q_2 | X = x) \quad (6.14)$$

for various Q_1, Q_2, x , and y .

We begin with the case of discrete random variables.

Let $V = \{y_1, y_2, \dots\}$ be the set of all possible values of Y . Then the event $\{Y = y\}$ appearing as the condition in (6.14) has positive probability only if $y \in V$. Here we can use the theory developed in Chapter 4, where we defined the conditional probability $P(A|B)$ of event A given that the event B occurred as $P(A|B) = P(A \cap B)/P(B)$, provided that $P(B) > 0$. If $P(B) = 0$, the probability $P(A|B)$ was left undefined. It is clear that to evaluate (6.14) in the case of discrete random variables, we do not need any new concepts. If the condition has probability zero (i.e., if $P\{Y = y\} = 0$), we leave $P\{X \in Q | Y = y\}$ undefined. For $y_j \in V$, we have $P\{Y = y_j\} > 0$, and

$$P(X \in Q | Y = y_j) = \frac{P(X \in Q, Y = y_j)}{P(Y = y_j)}. \quad (6.15)$$

Let $U = \{x_1, x_2, \dots\}$ be the set of possible values of X . If we write $p(x_i, y_j)$ or p_{ij} for $P\{X = x_i, Y = y_j\}$, the denominator in (6.15) is the marginal probability $P\{Y = y_j\} = \sum_i p_{ij} = p_{+j}$, and therefore

$$P\{X \in Q_1 | Y = y_j\} = \frac{\sum_{i: x_i \in Q_1} p_{ij}}{p_{+j}}.$$

An analogous formula with the role of X and Y interchanged is

$$P\{Y \in Q_2 | X = x_i\} = \frac{\sum_{j: y_j \in Q_2} p_{ij}}{p_{i+}}.$$

■ EXAMPLE 6.12

Let the experiment consist of two tosses of a die. We want to find conditional probabilities of the result of the first toss given the absolute difference of two tosses, and conversely, the conditional probabilities for the absolute difference given the results of the first toss. Accordingly, we let X_1 and X_2 denote the result of the first and the second toss, respectively, and put $Z = |X_1 - X_2|$.

The sample space $\mathcal{S}$ is naturally represented by the cells of the following table (where the values of Z are written in the cells):

X_1/X_2	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

The joint distribution of X_1 and Z is obtained by simple counting:

X_1/Z	0	1	2	3	4	5	X_1
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{6}$
2	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	$\frac{1}{6}$
3	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	$\frac{1}{6}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	$\frac{1}{6}$
5	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	$\frac{1}{6}$
6	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{6}$
Z	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$	1

We can now answer the various questions concerning conditional probabilities. For instance,

$$P(X_1 \text{ odd} | Z = 5) = \frac{P\{X_1 = 1, Z = 5\}}{P\{Z = 5\}} = \frac{\frac{1}{36}}{\frac{2}{36}} = \frac{1}{2},$$

while

$$P(Z > 2 | X_1 = 5) = \frac{P\{Z = 3, X_1 = 5\}}{P\{X_1 = 5\}} + \frac{P\{Z = 4, X_1 = 5\}}{P\{X_1 = 5\}} = \frac{1}{3}.$$

In the case of continuous random variables, determining a quantity such as, for example, $P\{X \in A | Y = y\}$ cannot rely upon the concepts of Chapter 4, since the conditioning event $\{Y = y\}$ has probability zero. We therefore *define* the conditional density and then verify that it has all the necessary properties.

Definition 6.3.1 The conditional densities g_{12} and g_{21} are defined by

$$g_{12}(x|y) = \frac{f(x, y)}{f_2(y)} \quad (6.16)$$

provided that $f_2(y) > 0$, and

$$g_{21}(y|x) = \frac{f(x, y)}{f_1(x)} \quad (6.17)$$

provided that $f_1(x) > 0$, where $f(x, y)$ is the joint density of (X, Y) and f_1 and f_2 are the marginal densities of X and Y , respectively. $\square$

If the denominators in (6.16) or (6.17) are zero, the left-hand sides remain undefined. To check that formulas (6.16) and (6.17) define densities, observe first that both functions, which are regarded as functions of the *first* variable (i.e., x in $g_{12}(x|y)$ and y in $g_{21}(y|x)$), are nonnegative. Moreover, we have

$$\int_{-\infty}^{+\infty} g_{12}(x|y) dx = \int_{-\infty}^{+\infty} \frac{f(x, y)}{f_2(y)} dx = \frac{1}{f_2(y)} \int_{-\infty}^{+\infty} f(x, y) dx = \frac{f_2(y)}{f_2(y)} = 1$$

and similarly for g_{21} . To justify the definition on “semantic” grounds, assume for simplicity that $f(x, y)$, and hence also $f_2(y)$, are continuous, and let us consider, for some $h > 0$, the probability $P\{X \in A | y \leq Y \leq y + h\}$. For small h , we have

$$\begin{aligned} P\{X \in A | y \leq Y \leq y + h\} &= \frac{P\{X \in A, y \leq Y \leq y + h\}}{P\{y \leq Y \leq y + h\}} \\ &= \frac{\int_A \int_y^{y+h} f(x, u) du dx}{\int_y^{y+h} f_2(u) du} \approx \frac{\int_A f(x, y) h dx}{f_2(y) h} \\ &= \int_A g_{12}(x|y) dx. \end{aligned}$$

This shows that $g_{12}(x|y)$ is a well-defined density of X , given that $Y = y$. A geometric interpretation of conditional density $g_{12}(x|y)$ is a section of surface $f(x, y)$ for $Y = y$ normalized by a constant so that it integrates to 1.

■ EXAMPLE 6.13

Let the joint density of (X, Y) be given by

$$f(x, y) = \begin{cases} cxy^2 & \text{if } 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Thus, the density is positive on the triangle with the vertices $(1, 1)$, $(1, 0)$, and $(0, 1)$; hence, X and Y are dependent (by the criterion for dependence of continuous random variables from Section 6.2). The marginal densities are as follows (we do not need to determine the numerical value of c ; it will cancel in densities g_{12} and g_{21}): If $0 < x < 1$, then

$$\begin{aligned} f_1(x) &= \int_{-\infty}^{+\infty} f(x, y) dy = \int_{1-x}^1 cxy^2 dy = cx \left. \frac{y^3}{3} \right|_{1-x}^1 \\ &= \frac{c}{3}(3x^2 - 3x^3 + x^4). \end{aligned}$$

Similarly, for $0 < y < 1$,

$$f_2(y) = \int_{1-y}^1 cxy^2 dx = cy^2 \left. \frac{x^2}{2} \right|_{1-y}^1 = \frac{c}{2}y^3(2-y).$$

Thus

$$g_{12}(x|y) = \frac{f(x,y)}{f_2(y)} = \frac{cxy^2}{(c/2)y^3(2-y)} = \frac{2x}{y(2-y)};$$

hence,

$$g_{12}(x|y) = \begin{cases} \frac{2x}{2y-y^2} & \text{for } 1-y < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

For instance, if $Y = 0.5$, then for $0.5 < x < 1$,

$$g_{12}(x|0.5) = g_{12}(x|Y = 0.5) = \frac{8}{3}x.$$

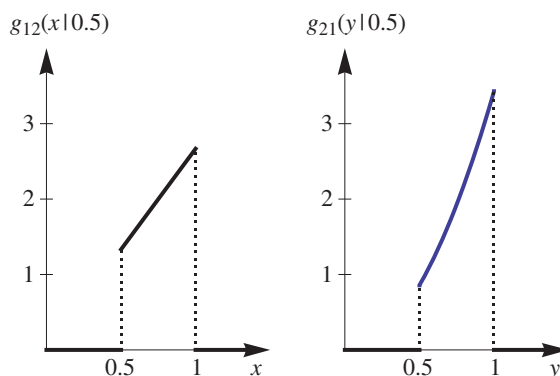


Figure 6.10 Conditional densities.

Similarly, for $1-x < y < 1$,

$$g_{21}(y|x) = \frac{f(x,y)}{f_1(x)} = \frac{cxy^2}{(c/3)(3x^2 - 3x^3 + x^4)} = \frac{3y^2}{x(3 - 3x + x^2)}$$

and equals 0 otherwise. Thus, if $X = 0.5$, for $0.5 < y < 1$ the density $g_{21}(y|0.5) = 24y^2/7$ (see Figure 6.10).

In Chapter 4, we repeatedly stressed the fact that the formula $P(A|B) = P(A \cap B)/P(B)$ can be used to determine the probability $P(A \cap B)$ of two events occurring jointly, given the probability of one of those events and the appropriate conditional probability, such as $P(B)$ and $P(A|B)$. The same situation is true in the case of conditional distributions in a continuous, discrete, or mixed case; given one marginal distribution and one conditional distribution, one can determine the joint distribution and hence also the marginal distribution of the other variable. We will illustrate this technique by three examples.

EXAMPLE 6.14

An animal lays a certain number X of eggs, where X is random and has the Poisson distribution

$$P\{X = n\} = \frac{\lambda^n}{n!} e^{-\lambda}, \quad n = 0, 1, 2, \dots$$

(see Example 5.9). Each egg hatches with a probability p , independent of the hatching of other eggs. Determine the distribution of Y = the number of eggs that hatch.

SOLUTION. Here the randomness of Y has two sources: first, the number X of eggs laid is random (varies from animal to animal), and second, even for animals that laid the same number of eggs, the randomness in the process of hatching may make the numbers of offspring different.

Our solution strategy is as follows: We are given the distribution of X . The two assumptions of the problem will allow us to determine the conditional distribution of Y given X . These assumptions allow us to determine the joint distribution of (X, Y) , and the distribution of Y will be obtained as marginal from the joint distribution of (X, Y) .

We now need to determine the conditional distribution of Y given X , that is, $P\{Y = j|X = n\}$. First, it is clear that $0 \leq Y \leq X$ (the number of eggs that hatch is nonnegative and cannot exceed the number of eggs laid). We assume that eggs hatch with the same probability and independently. Thus, Y must have binomial distribution if we regard the process of incubation as an “experiment over an egg” with “success” identified with hatching. Therefore,

$$P\{Y = j|X = n\} = \binom{n}{j} p^j (1-p)^{n-j}, \quad j = 0, 1, \dots, n.$$

By formula (6.15) the joint distribution of (X, Y) is

$$\begin{aligned} P\{X = n, Y = j\} &= P\{Y = j|X = n\} \times P\{X = n\} \\ &= \binom{n}{j} p^j (1-p)^{n-j} \times \frac{\lambda^n}{n!} e^{-\lambda}, \end{aligned}$$

where $n = 0, 1, 2, \dots$ and $j = 0, 1, \dots, n$ (for all other values of n and j , the joint probability is zero).

We will next find the marginal probability $P\{Y = j\}$ for $j = 0, 1, 2, \dots$. Clearly, for $X \geq Y$, we have $n \geq j$. Hence,

$$\begin{aligned} P\{Y = j\} &= \sum_{n=j}^{\infty} P\{X = n, Y = j\} = \sum_{n=j}^{\infty} \frac{\lambda^n}{n!} e^{-\lambda} \binom{n}{j} p^j (1-p)^{n-j} \\ &= \sum_{n=j}^{\infty} \frac{\lambda^n}{n!} e^{-\lambda} \frac{n!}{j!(n-j)!} p^j (1-p)^{n-j} \\ &= \frac{(\lambda p)^j}{j!} e^{-\lambda} \sum_{n=j}^{\infty} \frac{[\lambda(1-p)]^{n-j}}{(n-j)!} \\ &= \frac{(\lambda p)^j}{j!} e^{-\lambda} \sum_{\nu=0}^{\infty} \frac{[\lambda(1-p)]^{\nu}}{\nu!} = \frac{(\lambda p)^j}{j!} e^{-\lambda} e^{\lambda(1-p)} \\ &= \frac{(\lambda p)^j}{j!} e^{-\lambda p}. \end{aligned}$$

The marginal distribution of Y is again Poisson, except that the parameter has changed from λ to λp .

The process that leads from X to Y in this example is sometimes called *binomial thinning*. One can visualize it, in general, as a random process that gives the value of X (by assumption, an integer), with some distribution. We can think of X as the number of objects of some kind that are produced, a number of elements sampled, a number of events of some kind that occur, and so on. The process of thinning causes some of the X objects (events, etc.) to disappear (not be counted), due to a certain random process of selection. For instance, some of the X objects produced are defective, some of the X elements sampled are not acceptable or have certain other characteristic, some of the events are unobservable, and so on. Such a process of elimination of some X s and the selection of others is called *thinning*. We say that the process of thinning is binomial if the inclusions of X -elements as Y -elements are independent and occur with the same probability. The present example shows that binomial thinning of Poisson random variables leads again to Poisson random variables, with appropriately modified parameters.

We will now consider similar situation for continuous distributions.

■ EXAMPLE 6.15

A point X is chosen at random from the interval $[A, B]$ according to the uniform distribution (see Example 5.14), and then a point Y is chosen at random, again with uniform distribution, from the interval $[X, B]$. Find the marginal distribution of Y .

SOLUTION. Since

$$f_1(x) = \begin{cases} \frac{1}{B-A} & \text{for } A \leq x \leq B \\ 0 & \text{otherwise} \end{cases}$$

and

$$g_{21}(y|x) = \begin{cases} \frac{1}{B-x} & \text{for } x \leq y \leq B \\ 0 & \text{otherwise,} \end{cases}$$

we have

$$f(x, y) = f_1(x) g_{21}(y|x) = \begin{cases} \frac{1}{B-A} \times \frac{1}{B-x} & \text{for } A \leq x \leq y \leq B \\ 0 & \text{otherwise.} \end{cases}$$

Consequently, for fixed y ($A \leq y \leq B$),

$$f_2(y) = \int_A^y \frac{1}{B-A} \times \frac{1}{B-x} dx = \frac{1}{B-A} \log \frac{B-A}{B-y}.$$

The values close to B are more likely than those close to A ; the density of Y is in fact unbounded as we approach the upper boundary B of the range of Y .

■ EXAMPLE 6.16

Finally, we will consider the case of a mixed distribution, with X being continuous and Y being discrete.

Assume that Y takes one of $n \geq 2$ integer values with $P\{Y = i\} = p_i$ and $p_1 + p_2 + \cdots + p_n = 1$. In the simplest case, $n = 2$, we have

$$p_1 = P\{Y = 1\} = p, \quad p_2 = P\{Y = 2\} = 1 - p.$$

Next, assume that for a given $Y = i$, the random variable X has a continuous distribution with density $\varphi_i(x)$. Thus, we have

$$P\{a \leq X \leq b, Y = i\} = p_i \int_a^b \varphi_i(x) dx.$$

To find the marginal distribution of X , we can use the formula for total probability (4.9) from Chapter 4, with events $\{Y = i\}$ as the partition. Then

$$\begin{aligned} P\{a \leq X \leq b\} &= \sum_{i=1}^n P\{a \leq X \leq b | Y = i\} \times P\{Y = i\} \\ &= \sum_{i=1}^n p_i \int_a^b \varphi_i(x) dx = \int_a^b \sum_{i=1}^n p_i \varphi_i(x) dx. \end{aligned} \quad (6.18)$$

It follows that X is a continuous random variable with the density

$$f_X(x) = \sum_{i=1}^n p_i \varphi_i(x),$$

called the *mixture* of densities φ_i with mixing coefficients p_i .

If we have only two values of Y , and therefore only two densities φ_1 and φ_2 of X , then

$$f_X(x) = p\varphi_1(x) + (1 - p)\varphi_2(x).$$

The formulas above remain valid if Y assumes one of infinitely many values $1, 2, \dots$, with probabilities $p_1, p_2, \dots$ such that we have $\sum p_i = 1$. The only potential source of trouble is the interchange of integration and summation in the last step in (6.18). But this interchange is permissible because the terms of the sum are all nonnegative.

PROBLEMS

- 6.3.1** Suppose that three cards are drawn without replacement from an ordinary deck. Let X be the number of aces among the cards drawn and Y be the number of red cards among them. Find: **(i)** The joint distribution of (X, Y) . **(ii)** The conditional distribution of the number of aces if it is known that all three cards selected are red.
- 6.3.2** Let X and Y have the joint density $f(x, y) = \lambda^2 e^{-\lambda y}$ for $0 \leq x \leq y$ and $f(x, y) = 0$ otherwise. Find: **(i)** The joint cdf of (X, Y) . **(ii)** The marginal densities of X and Y . **(iii)** The conditional density of Y given X .
- 6.3.3** Refer to Problem 6.2.15 and find: **(i)** The conditional density of X given Y and of Y given X . **(ii)** Probability that a randomly selected customer spent more than 30 minutes at the counter if he was served less than 15 minutes. **(iii)** Probability that customer's waiting time was less than 10 minutes if he was served between 10 and 20 minutes.
- 6.3.4** Let variables X and Y be independent, each with $U[0, 1]$ distribution. Find: **(i)** $P(X + Y \leq 0.5 \mid X = 0.25)$. **(ii)** $P(X + Y \leq 0.5 \mid X \geq 0.25)$. **(iii)** $P(X \geq Y \mid Y \geq 0.5)$.

6.3.5 Let X and Y have joint density of the form

$$f(x, y) = \begin{cases} A(y - x)^\alpha & \text{for } 0 \leq x < y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find: **(i)** The values of α such that f can be a density function. **(ii)** The value of A for α specified in part (i). **(iii)** The marginal densities of X and Y . **(iv)** The conditional densities $g_{12}(x|y)$ and $g_{21}(y|x)$.

6.3.6 The phrase “A stick is broken at random into three pieces” can be interpreted in several ways. Let us identify the stick with interval $[0, 1]$ and let $0 < X < Y < 1$ be the breaking points. Some of the possible ways of generating X, Y are as follows: **(i)** A point (U, V) is chosen from the unit square with uniform distribution, and we let $X = \min(U, V)$ and $Y = \max(U, V)$. **(ii)** The point U is chosen from $[0, 1]$ with uniform distribution. If $U < 1/2$, then V is chosen with uniform distribution from $[U, 1]$, whereas if $U \geq 1/2$, then V is chosen with uniform distribution on $[0, U]$. Then (X, Y) are defined as in (i). **(iii)** X is chosen from $[0, 1]$ according to the uniform distribution, and then Y is chosen with uniform distribution on $[X, 1]$. **(iv)** U is chosen with uniform distribution on $[0, 1]$. Next, one of the intervals $[0, U]$ or $[U, 1]$ is chosen at random, with probability U and $1 - U$, respectively. Then V is chosen with uniform distribution from the chosen interval, and again, $X = \min(U, V)$ and $Y = \max(U, V)$. In each of the cases, (i)–(iv) find the joint density of (X, Y) and the marginal densities of X and Y . Which of the ways (i)–(iv) are equivalent?

6.3.7 A fast-food restaurant has a dining room and a drive-thru window. Let X and Y be the fractions of time (during a working day) when the dining room (X) and the drive-thru window (Y) are busy. Assuming that the joint density of (X, Y) is $f(x, y) = k(2x^2 + y^2)$ for $0 \leq x \leq 1, 0 \leq y \leq 1$ and $f(x, y) = 0$ otherwise, find: **(i)** k . **(ii)** The marginal densities of X and Y . **(iii)** The probability that the drive-thru window will be busy more than 75% of the time on a day when the dining room is empty less than 10% of the time. **(iv)** Do you find anything disturbing in this problem? If so, explain.

6.4 BIVARIATE TRANSFORMATIONS

In Chapter 5, we considered transformations of single random variables. Here we will consider functions of two continuous random variables X and Y with the joint density $f(x, y)$ with support C (so that $f(x, y) > 0$ on C and $f(x, y) = 0$ outside C). The objective is to find the density of $Z = \varphi(X, Y)$, where φ is a differentiable function of two real arguments. By far the simplest here is the cdf technique, introduced in Section 6.4 for the case of a single variable. It may also be applied in the multivariate case if we can obtain $P\{Z \leq z\} = P\{\varphi(X, Y) \leq z\}$ in a closed form as a function of z . Density can then be obtained by differentiation. This method works especially well for $\varphi(X, Y) = \max(X, Y)$ and $\varphi(X, Y) = \min(X, Y)$, when X and Y are independent. For example, $P\{\max(X, Y) \leq z\} = P\{X \leq z, Y \leq z\} = F_X(z)F_Y(z)$; hence, the density of $Z = \max(X, Y)$ is $f_X(z)F_Y(z) + F_X(z)f_Y(z)$.

Now, we will present a technique that may be applied to a wider class of cases. It will be given in the form of an algorithm, and its use will be illustrated by several examples. A formal proof will not be given, since the algorithm is in fact based on a change of variables in two-dimensional integrals, which can be found in advanced-level calculus texts.

Determination of densities of bivariate (and multivariate) transformations is typically regarded by students as challenging. We hope that by presenting it as a purely mechanical procedure—which it largely is—we will alleviate, or perhaps eliminate, the terror. The algorithm is as follows:

1. Choose a “companion” function, say $w = \eta(x, y)$, such that the pair of equations

$$z = \varphi(x, y) \quad \text{and} \quad w = \eta(x, y) \quad (6.19)$$

can be solved, leading to

$$x = \alpha(z, w) \quad \text{and} \quad y = \beta(z, w). \quad (6.20)$$

2. Determine the image D of the support C of density $f(x, y)$ in the (z, w) plane under transformation (6.19).
 3. Find the Jacobian of transformation (6.19), that is, the determinant

$$J = \begin{vmatrix} \frac{\partial \alpha}{\partial z} & \frac{\partial \alpha}{\partial w} \\ \frac{\partial \beta}{\partial z} & \frac{\partial \beta}{\partial w} \end{vmatrix}. \quad (6.21)$$

4. Determine the joint density $g(z, w)$, of random variables $Z = \varphi(X, Y)$, $W = \eta(X, Y)$, given by the formula

$$g(z, w) = \begin{cases} f(\alpha(z, w), \beta(z, w))|J| & \text{for } (z, w) \in D \\ 0 & \text{for } (z, w) \notin D. \end{cases}$$

5. Compute the density of $Z = \varphi(X, Y)$ as the marginal density of $g(z, w)$:

$$g_Z(z) = \int_{-\infty}^{+\infty} g(z, w) dw = \int_{D_z} f(\alpha(z, w), \beta(z, w))|J|dw,$$

where $D_z = \{w : (z, w) \in D\}$.

Out of these five steps, only step 1 requires some moderate amount of thinking (or, at least some experience). The reason is that the choice of the companion transformation η affects all subsequent steps, making the calculations easy, difficult, or perhaps even impossible.

■ EXAMPLE 6.17 Sum of Random Variables

The operation of addition of random variables appears so often that it is worthwhile to derive general formulas here. For $z = \varphi(x, y) = x + y$, we have a possible choice of a companion transformation, $w = \eta(x, y) = x$. Hence, the inverse transformation (6.20) is

$$x = w \quad \text{and} \quad y = z - w,$$

so that $\alpha(z, w) = w$, and $\beta(z, w) = z - w$. Thus,

$$J = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix}$$

and $|J| = 1$. If (X, Y) have joint density $f(x, y)$, then the joint density of (Z, W) is $f(w, z - w)$. The density of Z is

$$g_Z(z) = \int_{-\infty}^{+\infty} f(w, z - w) dw. \quad (6.22)$$

Now, we have only to determine the effective limits of integration, that is, the set of values w (for given z) at which the integrand is positive.

■ EXAMPLE 6.18 Sum of Exponential Random Variables

When X and Y are independent, and both have exponential distributions with the same parameter λ , their joint density is

$$f(x, y) = \begin{cases} \alpha^2 e^{-\alpha(x+y)} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise.} \end{cases} \quad (6.23)$$

Consequently, the density $g_Z(z)$, as given by (6.22), is

$$g_Z(z) = \int_{-\infty}^{+\infty} f(w, z-w) dw = \int_{D_z} \alpha^2 e^{-\alpha z} dw,$$

where D_z is the set $\{w : w \geq 0, z-w \geq 0\}$. Since we must have $z > 0$ and $0 \leq w \leq z$,

$$g_Z(z) = \alpha^2 e^{-\alpha z} \int_0^z dw = \alpha^2 z e^{-\alpha z}, z \geq 0 \quad (6.24)$$

and $g_Z(z) = 0$ for $z < 0$. We recognize the density (6.24) as a gamma density with parameters 2 and α (see Example 5.27).

■ EXAMPLE 6.19 Sum of Two Uniform Random Variables

Let two independent variables X and Y both have $U(0, 1)$ distributions. Their joint density is then

$$f(x, y) = \begin{cases} 1 & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Formula (6.22) gives the density of variable $Z = X + Y$ as

$$g_Z(z) = \int_{D_z} 1 dw, \quad (6.25)$$

where $D_z = \{w : 0 < w < 1, 0 \leq z - w \leq 1\}$. Inequalities defining D_z , $0 < w < 1$ and $z - 1 \leq w \leq z$, can be written as

$$\max(0, z-1) \leq w \leq \min(z, 1).$$

We must have $0 \leq z \leq 2$, otherwise D_z is empty. For $0 \leq z \leq 1$, the set D_z is the interval $0 \leq w \leq z$, while for $1 \leq z \leq 2$, we have the interval $z-1 \leq w \leq 1$. Consequently (6.25) gives (see Figure 6.11)

$$g_Z(z) = \begin{cases} z & \text{for } 0 \leq z \leq 1 \\ 2-z & \text{for } 1 \leq z \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

This distribution is called *triangular*.

■ EXAMPLE 6.20

Formula (6.22) was obtained by taking a specific companion transformation $w = \eta(x, y) = x$. There is no compelling reason for this choice. We could, of course, have chosen $w = \eta(x, y) = y$, getting a very similar formula. However, we could also have chosen something more fancy, for example, $w = \eta(x, y) = x/(x+5y)$.

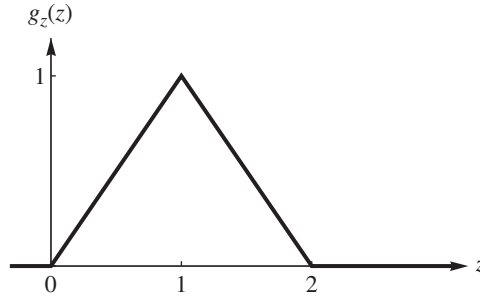


Figure 6.11 Triangular density.

The system of equations

$$z = x + y, \quad w = \frac{x}{x + 5y},$$

has the solution

$$x = \frac{5wz}{1 + 4w}, \quad y = z \frac{1 - w}{1 + 4w}.$$

Consequently, the Jacobian equals

$$J = \begin{vmatrix} \frac{5w}{1 + 4w} & \frac{5z}{(1 + 4w)^2} \\ \frac{1 - w}{1 + 4w} & \frac{-5z}{(1 + 4w)^2} \end{vmatrix} = -\frac{5z}{(1 + 4w)^2}.$$

Thus, the density of the sum $Z = X + Y$ can be obtained as the integral

$$g_Z(z) = \int_{-\infty}^{+\infty} f\left(\frac{5wz}{1 + 4w}, z \frac{1 - w}{1 + 4w}\right) \left| \frac{5z}{(1 + 4w)^2} \right| dw, \quad (6.26)$$

where again the effective range of integration depends on the support C of the joint density f .

This example is given here to show that there is no such thing as “the” formula for density of sum of random variables: Once the integration is carried out, (6.22) and (6.26) will both lead to the same final form of $g_Z(z)$. We cannot even say that (6.22) is simpler than (6.26), since the simplicity of integration depends here on the form of f and of its support C .

■ EXAMPLE 6.21 Product of Two Random Variables

We have $\varphi(x, y) = xy$, and we want to find the density of the random variable $Z = XY$.

Let us again choose the companion function $\eta(x, y) = x$ so that the system of equations (6.19) is $z = xy, w = x$, and its solution (6.20) is

$$x = \alpha(z, w) = w, \quad y = \beta(z, w) = \frac{z}{w}.$$

The Jacobian of this transformation is

$$J = \begin{vmatrix} 0 & 1 \\ \frac{1}{w} & -\frac{z}{w^2} \end{vmatrix} = -\frac{1}{w};$$

hence $|J| = 1/|w|$. The joint density of (Z, W) is now

$$f\left(w, \frac{z}{w}\right) \frac{1}{|w|};$$

and the density of Z is given by

$$g_Z(z) = \int_{-\infty}^{+\infty} f\left(w, \frac{z}{w}\right) \frac{1}{|w|} dw.$$

Again, the effective limits of integration depend on the support C of density $f(x, y)$ and, consequently, the sets D_z

We will now give an example that provides an algorithm of generating random variables with normal distribution.

Theorem 6.4.1 *If X and Y are independent, uniformly distributed on $(0, 1)$, then the random variables Z and W*

$$Z = \sqrt{-2 \log X} \sin(2\pi Y), \quad W = \sqrt{-2 \log X} \cos(2\pi Y),$$

are independent and each has the standard normal distribution.

Proof: Variables X and Y are independent, each with $U(0, 1)$ distribution. Their joint density is

$$f(x, y) = \begin{cases} 1 & \text{if } 0 < x < 1, \quad 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

For $z = \sqrt{-2 \log x} \sin(2\pi y)$ and $w = \sqrt{-2 \log x} \cos(2\pi y)$, we have $z^2 + w^2 = -2 \log x$, and $x = e^{-(z^2 + w^2)/2}$. On the other hand, $z/w = \tan(2\pi y)$, which gives

$$y = \frac{1}{2\pi} \arctan\left(\frac{z}{w}\right).$$

The Jacobian equals

$$J = \begin{vmatrix} -ze^{-(z^2 + w^2)/2} & -we^{-(z^2 + w^2)/2} \\ \frac{1}{w} & -z/w^2 \\ \frac{1}{2\pi(1 + z^2/w^2)} & \frac{-z/w^2}{2\pi(1 + z^2/w^2)} \end{vmatrix}, \quad (6.27)$$

which, after some algebra, reduces to

$$\frac{1}{\sqrt{2\pi}} e^{-z^2/2} \times \frac{1}{\sqrt{2\pi}} e^{-w^2/2}.$$

The unit square, equal to the support of $f(x, y)$, is mapped into the whole plane (z, w) . It follows that the joint density of (Z, W) , equal in this case to the Jacobian (6.27), is a product of two normal densities. $\square$

As already mentioned, the conditional densities given a specific value of a random variable cannot be calculated according to the principles of Chapter 4, since the conditioning event has probability zero. As a warning we present an example, seemingly paradoxical, where by following the rules as explained in this chapter, one obtains two different answers to the same question.

■ **EXAMPLE 6.22 Borel–Kolmogorov Paradox**

Assume that X and Y have joint density

$$f(x, y) = 4xy, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1,$$

so X and Y are independent, with the same marginal densities

$$f_X(t) = f_Y(t) = 2t, \quad 0 \leq t \leq 1.$$

We will try to determine the conditional density of X given the event $X = Y$. As we will see, the answer will depend on the representation of event $X = Y$, as $Y - X = 0$ or as $Y/X = 1$.

SOLUTION 1. We introduce new variables, $U = X, V = Y - X$. Then our problem becomes equivalent to finding the density of U , given $V = 0$. Thus, we have to find the joint density $h(u, v)$ of (U, V) and determine the marginal density $h_V(v)$. Our answer will be

$$\varphi_X(x|X=Y) = \frac{h(u, v)}{h_V(v)} \Big|_{v=0} = \frac{h(u, 0)}{h_V(0)}.$$

The transformation, $u = x, v = y - x$, has the inverse $x = u, y = u + v$, so the Jacobian is

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1.$$

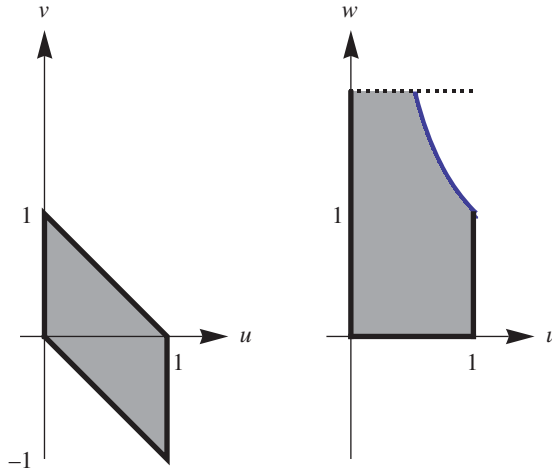


Figure 6.12 Supports of (U, V) and (U, W) .

Consequently, the joint density of (U, V) is $h(u, v) = 4u(u + v)$ for $0 \leq u \leq 1, -u \leq v \leq 1 - u$ (see Figure 6.12). The marginal density of V is

$$h_V(v) = \int_0^{1-v} 4u(u + v) du = 4 \left[\frac{(1-v)^3}{3} + v \frac{(1-v)^2}{2} \right] = \frac{2}{3}(1-v)^2(2+v),$$

for $0 \leq v \leq 1$, while for $-1 \leq v \leq 0$ we have

$$h_V(v) = \int_{-v}^1 4u(u+v) du = 4 \left[\frac{1}{3} + \frac{v}{2} - \frac{v^3}{6} \right] = -\frac{2}{3}(v+1)^2(v-2).$$

Both formulas give $h_V(0) = 4/3$. Thus, as the conditional density of U at $V = 0$, we obtain

$$\frac{h(u, 0)}{h_V(0)} = \frac{4u^2}{\frac{4}{3}} = 3u^2, \quad 0 \leq u \leq 1.$$

SOLUTION 2 We introduce the transformation $U = X, W = Y/X$. Then the conditional density of X , given $X = Y$, is the same as the conditional density of U , given $W = 1$.

The transformation $u = x, w = y/x$ has an inverse $x = u, y = uw$, and the Jacobian equals

$$J = \begin{vmatrix} 1 & 0 \\ w & u \end{vmatrix} = u.$$

Consequently, the joint density of (U, W) is $g(u, w) = 4u^3w$ for $0 \leq u \leq 1, 0 \leq w \leq 1/u$ (see Figure 6.12). The marginal density $g_W(w)$ is

$$g_W(w) = \int_0^1 4u^3w du = w \quad \text{for } 0 \leq w \leq 1,$$

while for $w \geq 1$ we have

$$g_W(w) = \int_0^{1/w} 4u^3w du = \frac{1}{w^3}.$$

For $w = 1$ we have $g_W(1) = 1$, and therefore the conditional density equals

$$\frac{g(u, 1)}{g_W(1)} = 4u^3, \quad 0 \leq u \leq 1.$$

Thus, we obtained two different solutions, and we may ask which of them—if either—is correct?

The formal answer is that both solutions are correct, and we can choose, for the density of X , given $X = Y$, any other function as well. This is simply because the conditional densities of a variable, given conditions with probability zero, can be modified arbitrarily on any set of conditions whose total probability is zero (in particular, on a single condition, or on finitely many conditions).

This answer is, however, not quite satisfactory. We also want to understand why the techniques of transformation lead to different answers. To explain this phenomenon, let us assume that we need the density of X given $X = Y$, but we cannot observe exactly whether the conditions holds. Specifically, consider two situations: (1) we can observe the difference $X - Y$, within some small error ϵ ; (2) we can observe the ratio Y/X , again within small error ϵ . The “natural” approach now is to use limiting passage, with $\epsilon \rightarrow 0$. For any fixed ϵ , the conditioning events $|X - Y| < \epsilon$ and $|Y/X - 1| < \epsilon$ are different (see Figure 6.13), and it is not surprising that the ratio Y/X favors larger values of u .

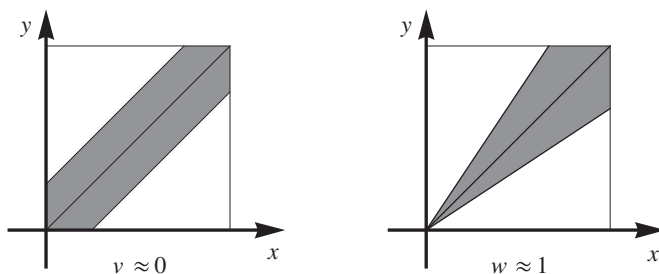


Figure 6.13 Approximations of two conditioning events.

PROBLEMS

6.4.1 Let X, Y be independent, each with a standard normal distribution. Find the distribution of: (i) $V = X/Y$. (ii) $U = (X - Y)^2/2$. (iii) $W = X/|Y|$.

6.4.2 Let random variables X_1, X_2 be independent and have both $\text{EXP}(\lambda)$ distribution. (i) Find the cdf and the density of $X_1 - X_2$. (ii) Find the cdf and the density of X_1/X_2 . (iii) Show that $Z_1 = X_1 + X_2$ and $Z_2 = X_1/(X_1 + X_2)$ are independent.

6.4.3 Let X and Y be independent random variables with densities

$$f_X(x) = c_1 x^{\alpha-1} e^{-x}, \quad f_Y(y) = c_2 y^{\beta-1} e^{-y}$$

for $x > 0, y > 0, \alpha > 0, \beta > 0$, and normalizing constants c_1, c_2 . Find the density of $W = X/(X + Y)$.

6.4.4 Independent random variables X and Y are uniformly distributed over intervals $(-1, 1)$, and $(0, 1)$, respectively. Find the joint distribution of variables $U = XY$ and $V = Y$. Determine the support of the density function.

6.4.5 Random variables (X, Y) have a joint density

$$f(x, y) = \begin{cases} k(ax + by) & \text{if } 0 < x < 2, \quad 0 < y < 2 \\ 0 & \text{otherwise,} \end{cases}$$

where $a > 0, b > 0$ are given constants. Find: (i) The value of k as a function of a and b . (ii) The density of variable $Z = 1/(Y + 1)^2$. (Hint: Express $F(t) = P\{Z \leq t\}$ in terms of the cdf of Y .)

6.4.6 Random variables X and Y have joint density $f(x, y) = cx$, $c > 0$, for $-x < y < x$, $0 < x < 1$, and zero otherwise. Find the density and the cdf of $W = X - Y$.

6.4.7 The joint distribution of random variables X and Y has density $f(x, y) = cx^2$, $c > 0$, for $0 < y < -x + 1, x > 0$ and 0 otherwise. Find the distribution of $W = X + Y$.

6.4.8 Let R be a nonnegative random variable with density $f(r)$. Let (X, Y) be a bivariate distribution obtained as follows: First, randomly choose a value of R , and then chose a value of U according to its $U(0, 1)$ distribution. Now, put

$$X = R \cos(2\pi U), \quad Y = R \sin(2\pi U). \quad (6.28)$$

Find: (i) The joint density of variables X and Y . (ii) $P\{XY > 0\}$. (iii) $P\{X > 0\}$. (iv) $P\{X^2 + Y^2 \leq t\}$. [Hint: For (i), find the joint density of (R, U) first and then use transformation (6.28).]

- 6.4.9** Let f be the joint density of a pair (X, Y) of random variables, and let a and b be two constants. Find the densities of (i) $aX + bY$. (ii) XY . (iii) X/Y . (iv) (U, V) , where $U = aX + b$, $V = cY + d$, and $ac \neq 0$.
- 6.4.10** Darts are thrown at a circular target. Let variables X and Y , coordinates of the point of impact be independent each having $N(0, \sigma^2)$ distribution. Find the density of D , the distance of the point of impact from the center of the target.
- 6.4.11** A current of I amperes following through a resistance of R ohms varies according to the probability distribution with density

$$f(i) = \begin{cases} 6i(1-i) & 0 < i < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the density of the power $W = I^2 R$ watts, if the resistance R varies independently of the current according to probability distribution with the density $g(r) = 2r$ for $0 < r < 1$, and zero otherwise.

6.5 MULTIDIMENSIONAL DISTRIBUTIONS

We will now extend the concepts introduced in the case of bivariate distributions to the case of multivariate (or multidimensional) distributions. The motivation for these concepts lies in the frequency of practical situations when the analysis concerns many random variables simultaneously.

The examples here are easy to find. First, often a description of the phenomenon studied (sampled objects, some process, effects of treatment, etc.) uses several attributes at once. Formally, we have several random variables, $X_1, \dots, X_n$, defined on a sample space, with X_i being the value of the i th attribute recorded for a given object. In such situations, a natural choice of sample space is to take the population of the objects under consideration, with probability P being generated by a specific scheme of selecting elements from the population.

In another context, we may think of repeated measurements of the same attribute so that X_i is the result of the i th measurement. Now, the probability P reflects possible dependence (or lack of it) in the measurement process.

Whatever the interpretation, formally, we have a vector of random variables

$$\mathbf{X} = \mathbf{X}^{(n)} = (X_1, \dots, X_n).$$

In the discrete case, the joint distribution of $\mathbf{X}$ consists of all probabilities of the form

$$P_{\mathbf{X}}(\mathbf{x}) = P\{\mathbf{X} = \mathbf{x}\} = P\{X_1 = x_1, \dots, X_n = x_n\},$$

where $\mathbf{x} = (x_1, \dots, x_n)$. Clearly, we must have $\sum_{\mathbf{x}} P_{\mathbf{X}}(\mathbf{x}) = 1$, the summation extended over all possible values $\mathbf{x} = (x_1, \dots, x_n)$ of vector $\mathbf{X}$.

In the continuous case, we have the joint density of the vector $\mathbf{X}$, in the form of a nonnegative function $f(\mathbf{x}) = f(x_1, \dots, x_n)$ of n variables such that

$$P\{\mathbf{X} \in Q\} = \underbrace{\int \cdots \int}_{\mathbf{x} \in Q} f(x_1, \dots, x_n) dx_1 \cdots dx_n$$

with

$$\underbrace{\int \cdots \int}_{R^n} f(x_1, \dots, x_n) dx_1 \cdots dx_n = 1.$$

The notions of marginal and conditional distributions and densities remain very much the same as in the bivariate case, except that the marginal distributions may now be themselves multivariate, and the same applies to conditional distribution, with the additional feature that the conditioning event may involve several random variables.

The simplicity of concepts can easily be obscured by confusing notation. While in any special case, there is seldom any danger of confusion (we know the meaning of the variables, and it is usually clear what is needed and what has to be done), the formulas covering the general cases may be confusing. We will use the following notation only presenting the theory; in Examples, we will try to simplify the notation, whenever possible.

First, in the case of marginal distribution, we need to specify the variables of interest. They form a subset of the variables $X_1, \dots, X_n$. Thus, we let $\mathbf{X} = (\mathbf{Y}, \mathbf{Z})$, where $\mathbf{Y}$ are the variables of interest and $\mathbf{Z}$ are the remaining variables. The question of ordering is irrelevant. For instance, if $\mathbf{X} = (X_1, X_2, X_3, X_4, X_5)$ and we are interested in the joint distribution of X_2 and X_5 , then $\mathbf{Y} = (X_2, X_5)$ and $\mathbf{Z} = (X_1, X_3, X_4)$. We let $\mathbf{y}$ and $\mathbf{z}$ denote the values of vectors $\mathbf{Y}$ and $\mathbf{Z}$ so that $\mathbf{x} = (\mathbf{y}, \mathbf{z})$ is the partitioning of $\mathbf{x}$ into the corresponding two subsets of coordinates. We now introduce the following definition:

Definition 6.5.1 In the discrete case, the *marginal distribution* of $\mathbf{Y}$ is given by

$$p_{\mathbf{Y}}(\mathbf{y}) = P\{\mathbf{Y} = \mathbf{y}\} = \sum_{\mathbf{z}} P\{\mathbf{Y} = \mathbf{y}, \mathbf{Z} = \mathbf{z}\}.$$

In the continuous case, the *marginal density* of $\mathbf{Y}$ is given by

$$f_{\mathbf{Y}}(\mathbf{y}) = \int \cdots \int_{\mathbf{z}} f(\mathbf{y}, \mathbf{z}) d\mathbf{z},$$

where the integrals represent the multiple integration over all variables X_i that are in vector $\mathbf{Z}$. $\square$

Before considering examples, let us look at the corresponding definitions for the case of conditional distributions. As in the bivariate case, the discrete distributions present no difficulty, and we will simply state that all formulas can be deduced starting from the definition of conditional probability for events, namely $P(A|B) = P(A \cap B)/P(B)$ if $P(B) > 0$. In the continuous case, we have to partition $\mathbf{X}$ into three components; so we write

$$\mathbf{X} = (\mathbf{Y}, \mathbf{Z}, \mathbf{W}),$$

where $\mathbf{Y}$ is the set of variables of interest, $\mathbf{Z}$ is the set of variables that will appear in the condition (whose values are assumed known), and $\mathbf{W}$ is the set (possibly empty) of variables that are neither in the condition nor of interest in the given instance. We need to define the conditional density of $\mathbf{Y}$ given $\mathbf{Z} = \mathbf{z}$.

Definition 6.5.2 The conditional density of $\mathbf{Y}$ given $\mathbf{Z} = \mathbf{z}$ is defined as

$$g_{\mathbf{Y}|\mathbf{Z}}(\mathbf{y}|\mathbf{z}) = \frac{f_{\mathbf{Y},\mathbf{Z}}(\mathbf{y}, \mathbf{z})}{f_{\mathbf{Z}}(\mathbf{z})} = \frac{\int f(\mathbf{y}, \mathbf{z}, \mathbf{w}) d\mathbf{w}}{\int \int f(\mathbf{y}, \mathbf{z}, \mathbf{w}) d\mathbf{y} d\mathbf{w}},$$

where the integral symbols represent multiple integration over the variables in vectors $\mathbf{w}$ and $\mathbf{y}$. $\square$

We illustrate that by several examples.

■ EXAMPLE 6.23 Trinomial Distribution

Let us consider an experiment in which the outcome can be classified into one of three exclusive and exhaustive categories. We let these categories be A, B , and C , and let α, β , and γ be their respective probabilities. Outcomes A, B , and C form a partition of all outcomes so that $\alpha + \beta + \gamma = 1$.

In the case of experiment on treatment effectiveness, A, B , and C could represent “improvement,” “relapse,” and “no change”; in cases of quality testing, the categories may be “acceptable,” “repairable,” “unacceptable and beyond repair,” and so on.

We can perform the experiment n times assuming that the repetitions are independent. Let X_1, X_2 stand for the counts of A and B among n repetitions (the count of C is $n - X_1 - X_2$). We then have the following theorem:

Theorem 6.5.1 *The joint distribution of $\mathbf{X} = (X_1, X_2)$ defined above is given by*

$$P\{X_1 = x_1, X_2 = x_2\} = \frac{n!}{x_1!x_2!(n - x_1 - x_2)!} \alpha^{x_1} \beta^{x_2} \gamma^{n-x_1-x_2}, \quad (6.29)$$

where x_1, x_2 are nonnegative integers such that $x_1 + x_2 \leq n$.

Proof: The probability of x_1 outcomes A , x_2 outcomes B , and $n - x_1 - x_2$ outcomes C in a specific order equals

$$P(A)^{x_1} P(B)^{x_2} P(C)^{n-x_1-x_2} = \alpha^{x_1} \beta^{x_2} \gamma^{n-x_1-x_2},$$

by assumption of independence. Considering all possible

$$\binom{n}{x_1, x_2, n - x_1 - x_2} = \frac{n!}{x_1!x_2!(n - x_1 - x_2)!}$$

orders (see Theorem 3.4.1), we obtain (6.5.1). □

■ EXAMPLE 6.24

Continuing Example 6.23, let us find the marginal distribution of X_2 . Formally, for $x_2 = 0, 1, \dots, n$, we have

$$p_{X_2}(x_2) = P\{X_2 = x_2\} = \sum_{x_1, x_3} P\{X_1 = x_1, X_2 = x_2, X_3 = x_3\}.$$

However, $x_3 = n - x_1 - x_2$, so

$$P\{X_1 = x_1, X_2 = x_2, X_3 = n - x_1 - x_2\} = P\{X_1 = x_1, X_2 = x_2\}.$$

Therefore, using (6.29), we write

$$\begin{aligned}
 p_{X_2}(x_2) &= \sum_{x_1=0}^{n-x_2} P\{X_1 = x_1, X_2 = x_2\} \\
 &= \sum_{x_1=0}^{n-x_2} \frac{n!}{x_1!x_2!(n-x_1-x_2)!} \alpha^{x_1} \beta^{x_2} (1-\alpha-\beta)^{n-x_1-x_2} \\
 &= \frac{n!}{x_2!(n-x_2)!} \beta^{x_2} (1-\beta)^{n-x_2} \\
 &\quad \times \sum_{x_1=0}^{n-x_2} \frac{(n-x_2)!}{x_1![(n-x_2)-x_1]!} \times \frac{\alpha^{x_1}}{(1-\beta)^{x_1}} \left(1 - \frac{\alpha}{1-\beta}\right)^{(n-x_2)-x_1} \\
 &= \binom{n}{x_2} \beta^{x_2} (1-\beta)^{n-x_2},
 \end{aligned}$$

since the last sum above reduces to

$$\sum_{x_1=0}^{n-x_2} \binom{n-x_2}{x_1} \pi^{x_1} (1-\pi)^{n-x_2-x_1} = 1,$$

where $\pi = \alpha/(1-\beta)$.

Thus, the marginal distribution of a variable in a trinomial distribution is binomial, as could have been expected. It is enough to interpret an occurrence of B as a success and the occurrence of anything else (A or C) as failure. So the probabilities of success and failure are β and $\alpha + \gamma = 1 - \beta$, respectively. The number of successes, X_2 , has a binomial distribution, as it should. In the general case, suppose that the outcomes of each experiment are categorized into $m+1$ classes, their probabilities being $\pi_1, \pi_2, \dots, \pi_{m+1}$, with $\pi_1 + \pi_2 + \dots + \pi_{m+1} = 1$. If the experiment is then repeated independently n times, and X_i ($i = 1, \dots, m$) is the number of outcomes of category i among n outcomes, then for nonnegative integers $x_1, \dots, x_m$ with $x_1 + \dots + x_m \leq n$ we have

$$P\{X_1 = x_1, \dots, X_m = x_m\} = \frac{n!}{x_1! \dots x_m! x_{m+1}!} \pi_1^{x_1} \dots \pi_m^{x_m} \pi_{m+1}^{x_{m+1}},$$

where $x_{m+1} = n - (x_1 + \dots + x_m)$ and $\pi_{m+1} = 1 - (\pi_1 + \dots + \pi_m)$. This is the *multinomial* distribution.

Let now $\mathbf{Y} = (X_{i_1}, \dots, X_{i_k})$ be a subset of variables $\mathbf{X} = (X_1, \dots, X_m)$ with a multinomial distribution. Proceeding in the same way as in Example 6.24, we can show that for $\mathbf{y} = (y_{i_1}, \dots, y_{i_k})$ with coordinates y_{i_j} being nonnegative integers with $y_{i_1} + \dots + y_{i_k} \leq n$, we have

$$P\{Y_{i_1} = y_{i_1}, \dots, Y_{i_k} = y_{i_k}\} = \frac{n!}{y_{i_1}! \dots y_{i_k}! y_{i_{k+1}}!} \pi_{i_1}^{y_{i_1}} \dots \pi_{i_k}^{y_{i_k}} \pi_{i_{k+1}}^{y_{i_{k+1}}},$$

where $y_{i_{k+1}} = n - y_{i_1} - \dots - y_{i_k}$ and $\pi_{i_{k+1}} = 1 - \pi_{i_1} - \dots - \pi_{i_k}$. Thus, a marginal distribution in a multinomial distribution is again multinomial with lower dimensionality.

EXAMPLE 6.25 Model of Grinding

The basic model suggested for the process of grinding rock begins with a piece of rock that is randomly divided into two parts. Each of these two parts is again divided randomly, and so on, until in the n th “generation” we have 2^n pieces of rock. Assume that this process continues for a large number of generations.

With the model one can obtain a distribution, as $n \rightarrow \infty$, of sizes of rock. Naturally, with each generation of grinding the actual sizes of rock will tend to zero, so that the sizes of 2^n pieces in the n th generation will have to be multiplied by the approximate scaling factor.

The results allow us to predict what fraction of the initial mass of rock will be ground into gravel with sizes contained between specific limits a and b , and so on.

Instead of the whole process, we will study only the first two divisions and the resulting sizes. For simplicity, we assume that the initial size is 1, represented as interval $[0, 1]$. In the first division, a point X —with uniform distribution—partitions the unit interval into two pieces of length X and $1 - X$. In the next partition, length X is divided by point Y (with Y distributed uniformly between 0 and X), and the remainder is divided by point Z , distributed uniformly on $[X, 1]$. We therefore have four fragments (see Figure 6.14) of length

$$X_1 = Y, \quad X_2 = X - Y, \quad X_3 = Z - X, \quad X_4 = 1 - Z.$$

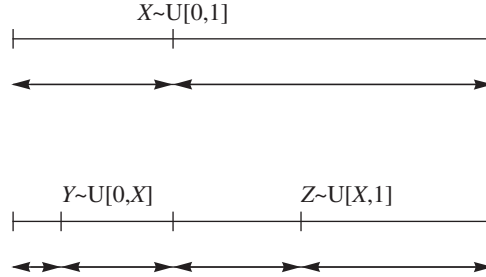


Figure 6.14 First two generations in the process of grinding.

We will start by deriving the joint trivariate density of (X, Y, Z) . By assumption, X has density

$$f_X(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Given $X = x$, variables Y and Z are independent with densities

$$g_{Y|X}(y|x) = \begin{cases} \frac{1}{x} & 0 \leq y \leq x \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad g_{Z|X}(z|x) = \begin{cases} \frac{1}{1-x} & x \leq z \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Thus, $g_{Y,Z|X}(y, z|x) = g_{Y|X}(y|x)g_{Z|X}(z|x)$, and the joint density is

$$f(x, y, z) = \begin{cases} \frac{1}{x(1-x)} & \text{for } 0 \leq y \leq x \leq z \leq 1 \\ 0 & \text{otherwise.} \end{cases} \quad (6.30)$$

The marginal joint density of Y and Z , for $0 \leq y \leq z \leq 1$ will now be

$$f_{Y,Z}(y, z) = \int_y^z \frac{dx}{x(1-x)} = \int_y^z \left(\frac{1}{x} + \frac{1}{1-x} \right) dx = \log \frac{z}{1-z} + \log \frac{1-y}{y},$$

so the marginal density of Y will be

$$f_Y(y) = \int_y^1 \left(\log \frac{z}{1-z} + \log \frac{1-y}{y} \right) dz = -\log y$$

for $0 < y < 1$. Finally, we obtain the conditional density of X given $Y = y$ and $Z = z$:

$$g_{X|Y,Z}(x|y, z) = \frac{f(x, y, z)}{f_{Y,Z}(y, z)} = \frac{\frac{1}{x(1-x)}}{\log \frac{z}{1-z} + \log \frac{1-y}{y}}$$

for $0 < y < x < z < 1$.

To derive the density of (X_1, X_2, X_3, X_4) , we need to derive only the density of (X_1, X_2, X_3) , since $X_1 + X_2 + X_3 + X_4 = 1$. This can be done by applying the transformation

$$x_1 = y, \quad x_2 = x - y, \quad x_3 = z - x \quad (6.31)$$

to the trivariate density (6.30).

The inverse transformation to (6.31) is

$$y = x_1, \quad x = x_1 + x_2, \quad z = x_1 + x_2 + x_3. \quad (6.32)$$

with Jacobian

$$J = \begin{vmatrix} \frac{\partial y}{\partial x_1} & \frac{\partial y}{\partial x_2} & \frac{\partial y}{\partial x_3} \\ \frac{\partial x}{\partial x_1} & \frac{\partial x}{\partial x_2} & \frac{\partial x}{\partial x_3} \\ \frac{\partial z}{\partial x_1} & \frac{\partial z}{\partial x_2} & \frac{\partial z}{\partial x_3} \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 1.$$

Thus, the joint density $g(x_1, x_2, x_3)$ of (X_1, X_2, X_3) is obtained from (6.30) by substituting (6.32), that is,

$$g(x_1, x_2, x_3) = \begin{cases} \frac{1}{(x_1 + x_2)(1 - x_1 - x_2)} & x_1 > 0, \ x_2 > 0, \ x_3 > 0, \\ & x_1 + x_2 + x_3 < 1 \\ 0 & \text{otherwise.} \end{cases}$$

PROBLEMS

6.5.1 Let variables X_1 and X_2 be independent, such that $P(X_i = -1) = P(X_i = 1) = 0.5$, $i = 1, 2$. Moreover, let $X_3 = X_1 X_2$. Show that variables X_1, X_2, X_3 are not independent but each two of them are independent (they are pairwise independent).

6.5.2 Let variables X_1, X_2 , and X_3 be independent, each having a $U(0, 1)$ distribution. Find: **(i)** The probability that exactly two of the three variables will be larger than 0.4. **(ii)** $P(X_1 + X_2 > X_3)$.

6.5.3 Two cards are drawn without replacement from an ordinary deck. Let X be the number of aces, Y be the number of red cards, and Z be the number of hearts. Find: **(i)** The joint distribution of (X, Y, Z) . **(ii)** $P\{Y = Z\}$. **(iii)** The conditional distribution of Z given $Y = y$.

6.5.4 Suppose that X_1, X_2, X_3 have joint density

$$f(x_1, x_2, x_3) = \begin{cases} x_1 x_2 x_3 & \text{for } 0 < x_i < c, \quad i = 1, 2, 3 \\ 0 & \text{otherwise.} \end{cases}$$

Find: **(i)** c . **(ii)** The joint density of (Y_1, Y_2, Y_3) , where $Y_1 = X_1, Y_2 = X_1 X_2$, and $Y_3 = X_1 X_2 X_3$.

6.5.5 Let X, Y , and Z have joint density $f(x, y, z) = c(x + y + z)$ for $0 < x < y < z < 1$. Find: **(i)** c . **(ii)** $P(X + Z > 1)$. **(iii)** $P(X + Z > 1 | Y = 0.5)$.

6.5.6 Assume that variables X, Y , and Z are jointly distributed with the density $f(x, y, z) = x + y^2 + z^5$ for $0 < x < 1, 0 < y < 1, 0 < z < 1$, and 0 otherwise. Find $P(Y < X | Z = 0.2^{1/5})$.

6.5.7 A lifetime of some electronic unit has density $f(t)$. Each time the unit fails, it is replaced by another one with the same lifetime distribution. Let X, Y , and Z be lifetimes of three consecutive units independently installed one after the other failed. Find: **(i)** $P(X < Y < Z)$. **(ii)** $P(X < Y < Z)$ if we know that $f(t) = 2te^{-t^2}$ for $t > 0$.

6.5.8 A can of Three-Bean-Salad contains beans of varieties A, B , and C (plus other ingredients which are of no concern for the problem). Let X, Y , and Z denote the relative weights of varieties A, B , and C in a randomly selected can (so that $X + Y + Z = 1$). Moreover, let the joint distribution of (X, Y) have the density $f(x, y) = kx^2y$ for $x > 0, y > 0, x + y < 1$, and $f(x, y) = 0$ otherwise. Find: **(i)** k . **(ii)** Probability $P(X > 0.5)$ that beans A take more than half of the total weight. **(iii)** $P(Z > 0.5)$. **(iv)** Probability that none of the three varieties of beans will take more than half of the weight. **(v)** The marginal density of (X, Z) and of Z .

6.5.9 Let four observations X_1, X_2, X_3, X_4 be independently selected from the same distribution with density $f(x) = e^{-x}$ for $x > 0$ and 0 otherwise. Find: **(i)** The probability that exactly one of these observations is less than 1. **(ii)** $P(X_1 + X_2 < X_3 + X_4)$.